(18)

Supreme Court, U.S.
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No. 00-1543

In The

Supreme Court of the United States

FESTO CORPORATION,

Petitioner,

v.

SHOKETSU KINZOKU KOGYO KABUSHIKI CO., LTD.,
a/k/a SMC CORPORATION and
SMC PNEUMATICS, INC.,

Respondents.

On Writ Of Certiorari
To The United States Court Of Appeals
For The Federal Circuit

JOINT APPENDIX
VOLUME I, PAGES I-1 TO 1-153

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RELEVANT DOCKET ENTRIES

United States District Court

Date	N.R.	PROCEEDINGS
Aug 4	1	Complaint, FILED, S/I CFI LR10 18A
Sep 26	3	D's ANSWER and COUNTER-CLAIMS, FILED. cs
Sep 30	5	D's Amended ANSWER and counter claims, FILED. cs
Oct 19	6	P's demand for trial by jury, FILED. cs
Nov 8	10	P Festo Corporation's Reply to D's Counterclaims, FILED. cs
1989		
Oct 2	39	*MAZZONE, D.J.*, Memorandum and Order, entered. Since Festo KG did not take any action itself to establish minimum contacts in Massachusetts and none of Festo US's activities can be deemed as performed for Festo KG, I find that the criteria of the Massachusetts long-arm statute are not satisfied. Accordingly, the motion of Festo KG to dismiss is allowed. cc/cl
1990		
Oct 24	46	Amended Complaint, Filed.
Nov 26	50	Pltf Reply to Dfts Cntrol in Dft's Response to Pltf's Amended Complaint filed cs
Nov	51	Pltf Demand a Trial by Jury on all issues, filed c/s.

U.S. District Court – Massachusetts (Boston)

1/31/92	95	Trial briefs by Festo Corp. (mcm) [1:88cv1814]
2/21/92	99	Trial briefs by Smc Pneumatics, Inc. filed cs. (dlw) [1:88cv1814]
2/26/92	104	Motion by Shoketsu Kinzoku Kog, Smc Pneumatics, Inc. for summary judgment mtn for stoll patent invalidity filed cs. (mcm) [1:88cv1814]
2/28/92	–	Status conference held Re; schedule for trial. D's reply trial Doc. filed in court under seal. Based on its review of the [trial] documents, the court has concluded that this case is inappropriate for trial to jury. The parties have ruled out settlement. Accordingly, the parties may elect on[e] of two options: 1. agree to a Special Master, or submit 3 names each and the court will select; 2. Show cause why a rule 706 expert should not be appointed. The parties have 20 days to report to the clerk their decision. (sad) [Entry date 04/08/92] [1:88cv1814]
2/28/92	138	Judge A. D. Mazzone. Clerk's Notes:, Trial is hereby conce[a]led. Ruling on all pending motions deferred pending appointment of special Master or Rule 706 Expert. . . . terminated deadlines (sad) [Entry date 04/08/92] [1:88cv1814]
3/20/92	117	Motion by Shoketu Kinzoku Kog, Smc Pneumatics, Inc. for summary judgment of non-infringement of the stoll patent

		filed cs. (mcm) [Entry date 03/23/92] [1:88cv1814]
3/20/92	118	Memorandum by Shoketsu Kinzoku Kog, Smc Pneumatics, Inc. in support of [117-1] motion for summary judgment of non-infringement of the stoll patent (mcm) [Entry date 03/23/92] [1:88cv1814]
3/20/92	119	Statement of Facts by Shoketsu Kinzoku Kog, Smc Pneumatics, Inc. Re: [117-1] motion for summary judgment of non-infringement of the stoll patent (mcm) [Entry date 03/23/92] [1:88cv1814]
3/24/92	122	Motion by Shoketsu Kinzoku Kog, Smc Pneumatics, Inc. for summary judgment of non-infring of the stoll and carroll patents by the one seal SMC Rodless Cylinder filed cs. (req for oral argument) (mcm) [1:88cv1814]
5/1/92	144	EMERGENCY/Motion by Festo Corp. to strike [122-1] motion for summary judgment of non-infring of the stoll and carroll patents by the one seal SMC Rodless Cylinder, [117-1] motion for summary judgmnt of non-infringement of the stoll patent, [104-1] motion for summary judgment mtn for stoll patent invalidity filed cs. REQ FOR ORAL ARGUMENT (mcm) [Entry date 05/06/92] [1:88cv1814]
5/7/92	149	Response by Festo Corp. in opposition to [117-1] motion for summary judgment of non-infringement of the stoll patent (mcm) [Entry date 05/14/92] [1:88cv1814]

5/21/92	166	Judge A. D. Mazzone. Order referring case to Herbert F. Schwartz as special master for the purpose of considering all claims presented in this case by the parties and for such other matter as may be referred to the Master by the Court such as discovery disputes. (mcm) [Entry date 06/02/92] [1:88cv1814]
5/21/92	166	Judge A. D. Mazzone. Order of reference entered for Alternative Dispute Resolution. Referred to: Herbert Schwartz for: alternative dispute (mcm) [Entry date 01/24/95] [1:88cv1814]
7/8/92	170	Ltr from Special Master H. Schwartz re: confirming ctrm 1110 for hearings on the following dates: September 14-18, 1992, September 21-15, 1992 and October 19-23, 1992. Arrangements for a court reporter will be made by Mr. Hoffman and Mr. Neustadt. (mcm) [1:88cv1814]
8/13/92	172	Order of Special Master Herbert F. Schwartz Entered (sad) [Entry date 08/24/92] [1:88cv1814]
4/27/93	176	Report of the Special Master Herbert F. Schwartz, Esq., Pursuant To Rule 53(e)(1) F.R.Civ.P., filed. (see file dkt #176 for further details) (mcm) [1:88cv1814]
5/11/93	178	Motion by Shoketsu Kinzoku Kog, Smc Pneumatics, Inc. for entry of summary judgment and direct verdict of Non-Infringement of the stoll patent FILED c.s. (jmr) [1:88cv1814]

5/18/93	184	Judge A. D. Mazzone. Order, Objections or motions with supporting memoranda for confirmation of Special Master's Report shall be filed and served by 6/9/93. Entered cc/cl (sad) [Entry date 06/08/93] [1:88cv1814]
6/9/93	185	Motion by Festo Corp. to vacate or modify those portions of the Report of the SM in accord w/pltf objection filed and to confirm the remaining portions of the report filed cs. (mcm) [1:88cv1814]
6/9/93	186	Response by Festo Corp. in opposition to the Report of the Special Master, pursuant to Rule 53(e)(3), FR Civ P, filed cs. (mcm) [1:88cv1814]
6/9/93	187	Motion by Festo Corp. for summary judgment of validity and a directed verdict of the reexamined Carroll Patent filed cs. (mcm) [1:88cv1814]
6/9/93	191	Motion by Festo Corp. for summary judgment and or a directed verdict of infringement of the reexamined Carroll Patent filed cs. (mcm) [1:88cv1814]
6/9/93	194	Motion by Festo Corp. for summary judgment and a directed verdict of VALIDITY OF THE STOLL PATENT filed cs. (mcm) [1:88cv1814]
6/9/93	198	Motion by Festo Corp. for summary judgment and directed verdict of INFRINGEMENT OF THE STOLL PATENT filed cs. (mcm) [1:88cv1814]
6/9/93	202	Reply by Smc Pneumatics, Inc. to response to the May 18, 1993 ORDER

CONCERNING THE SPECIAL MASTER'S REPORT, filed cs. (mcm) [1:88cv1814]

6/30/93 205 Response by Smc Pneumatics, Inc. in opposition to [198-1] (4) motions for summary judgment and directed verdict of INFRINGEMENT OF THE STOLL PATENT (mcm) [Entry date 07/07/93] [1:88cv1814]

1/25/94 – Case reassigned from Judge Mazzone to Judge Patti B. Saris upon appointment to the bench. Notice (1/18/94) sent to counsel. (mcm) [1:88cv1814]

2/3/94 209 Judge Patti B. Saris. Clerk's Notes: hearing on pending motions for SJ. Parties to consider trial by jury on liability, and to court on damages. Special Masters report will be redacted and submitted to the jury. Pltfs physics expert will be allowed; dft can depose physics expert. NOTED that Miamoto patent was not addressed in SM trial, but is the substance of SMC's counterclaim. Trial will be limited to one week per side. FPTC set for 5/6/94. Jury trial set for 5/16/94, 10-15 days. Settlement cnf held off the record. Parties to report to clerk by 2/11/94 re: status of settlement and if they want settlement cnf w/principals scheduled w/Sr. Judge Mazzone. Further briefing order to issue. (mcm) [Entry date 02/25/94] [1:88cv1814]

2/3/94 – Judge Patti B. Saris. Endorsed Order denying [178-1] motion for entry of summary judgment and direct verdict of Non-Infringement of the stoll patent

(mcm) [Entry date 02/25/94] [1:88cv1814]

2/3/94 – Judge Patti B. Saris. Endorsed Order denying [194-1] motion for summary judgment and a directed verdict of VALIDITY OF THE STOLL PATENT (mcm) [Entry date 02/25/94] [1:88cv1814]

2/3/94 – Judge Patti B. Saris. Endorsed Order denying [198-1] motion for summary judgment and directed verdict of INFRINGEMENT OF THE STOLL PATENT (mcm) [Entry date 02/25/94] [1:88cv1814]

2/18/94 208 Judge Patti B. Saris. Procedural Order, set settlement conference before Judge A. D. Mazzone for 10:00 3/16/94, set pretrial conference for 2:00 5/6/94 before Judge Patti B. Saris. For the reasons stated in court on 2/3/94, the parties mtns for s.j dkt #104,117,122,178,194 and 198 are DENIED. The parties mtns for s/j dkt # 122 and 187 are DENIED. Ruling is DEFERRED on the pltfs mtn for mtn for s/j and directed verdict of infringement on the reexamined Carroll patent dkt #191 pending receipt of supp briefing. by 2/25/94, the parties shall file supp memo on the following issues: new case on doctrine of equivalents (Carroll patent); literal infringement (Carroll patent); any other issues raised. Any responses to the above submissions shall be filed no later than March 4, 1994. (see file dkt #208 for further

details) (mcm) [Entry date 02/25/94] [1:88cv1814]

2/25/94 210 Pltf Festo Corp., Supplemental Memorandum Re: New case law on the Doctrine of Equivalents with respect to the Carroll Patent, filed cs. (mcm) [Entry date 02/28/94] [1:88cv1814]

2/25/94 211 SMC'S Memo concerning the absence of literal infringement of the Carroll Patent, filed. (mcm) [Entry date 02/28/94] [1:88cv1814]

2/25/94 212 Motion by Smc Pneumatics, Inc. for consideration of Special Master's recommendation filed cs. (mcm) [Entry date 02/28/94] [1:88cv1814]

3/4/94 213 Response by Festo Corp. to [212-1] motion for consideration of Special Master's recommendation, FILED. (c/s) (cmg) [Entry date 03/14/94] [1:88cv1814]

3/4/94 214 Response by Festo Corp. to SMC's Memorandum Concerning the Absence of Literal Infringement of the Carroll Patent, FILED. (c/s) (cmg) [Entry date 03/14/94] [1:88cv1814]

3/4/94 216 Response of SMC's to Festo's supplemental memo regarding new case law on the doctrine of equivalents, filed cs. (mcm) [Entry date 03/16/94] [1:88cv1814]

3/14/94 215 Transcript of proceedings for motion hearing held on 2/4/94 before Judge: Saris (mcm) [1:88cv1814]

3/30/94 – DFTS SUPPLEMENTAL Memorandum by Smc Pneumatics, Inc. in support of [178-1] motion for entry of summary judgment and direct verdict of Non-Infringement of the stoll patent (mcm) [Entry date 04/04/94] [1:88cv1814]

5/3/94 220 Pretrial memorandum by Festo Corp., filed. (mcm) [Entry date 05/04/94] [1:88cv1814]

5/10/94 222 Judge Patti B. Saris. Clerk's Notes: re: FINAL PRETRIAL CONFERENCE HELD, Trial date continued . . . set Jury trial for 5/31/94 Plaintiff h/f/updated trial document; dft will stand on previously-filed trial document. No motions in limine f. Trial to proceed w/o bifurcation. Counsel anticipate 10-15 trial days, incl. damages and break for liability deliberations (plaintiff's case infringement; dft's case invalaidity) Order of presentation: plaintiff's case first, dft's second. Counterclaim on Miamoto patent is waived except for attys. fees issue. Jury to decide willfulness; Court to decide attys fees. Submission of MASTER'S REPORT (issue briefed earlier by pltf): Counsel w/h/t/stip to fact findings in SM's report: redact rest. SM REPORT t/b/given to jury to read, but not keep. Witnesses w/n/b permitted to comment on SM REPORT. TEN JURORS t/b/empaneled; all will deliberate. FIVE PEREMPTORY CHALLENGES PER SIDE. Local counsel need not always be present. Motions

		in limine, propsed jury interrogs., proposed voir dire, jury instructions t/b/f/by 5/23/94 . . . Both sides to supp. experts by 5/31/94. Parties anticipate 10-15 days of trial (1 for opening/closing statements, 4 for pltf's liability case, 4 for dft's, 2 days per side for dmgs, plus break for deliberations on liability phase). Court will hold parties to this schedule. (tmc) [Entry date 05/18/94] [Edit date 06/22/94] [1:88cv1814]
6/22/94	261	Response by Festo Corp. in opposition to Special Master Report of May 21, 1993, (DOCUMENT FILED UNDER SEAL) filed. (scj) [Entry date 09/14/94] [Edit date 09/14/94] [1:88cv1814]
6/22/94	262	Reply by Festo Corp. to response to the May 18, 1993 Order concerning the Special Master's Report (DOCUMENT FILED UNDER SEAL), filed. (scj) [Entry date 09/14/94] [1:88cv1814]
6/24/94	224	Trial brief by Smc Pneumatics, Inc., filed. (scj) [Entry date 06/27/94] [1:88cv1814]
6/24/94	226	Jury instructions filed. (scj) [Entry date 06/27/94] [1:88cv1814]
6/27/94	230	Trial brief by Festo Corp., filed. (scj) [Entry date 06/28/94] [1:88cv1814]
6/28/94	231	SMC's further comment and objection concerning the Special Master's Report (scj) [Entry date 06/29/94] [1:88cv1814]
6/29/94	–	Pre-trial conference held. (scj) [Entry date 07/01/94] [1:88cv1814]

6/29/94	236	Judge Patti B. Saris. Clerk's Notes: re: Final Pretrial Conference [229-1] motion in limine to exclude testimony and evidence of Defendants SMC Corporation and SMC Pneumatics, Inc. taken under advisement Court Reporter: Cloonan . . . Issues for trial . . . 1) Carroll patent (validity) 2) Stoll patent (validity) 3) Infringemenet on Carroll patent granted to P by way of ST, opinion to issue, SJ opinion to issue. SJ to dfts on Bolt issued. For purposes of trial, both dfts. w/b/ collectively referred as SMC. Proposed voir dire verdict form covered. Jury empanelment: 10 jurors, 5 peremtories per side. to dft on Bolt Issue. SJ (scj) [Entry date 07/01/94] [1:88cv1814]
7/5/94	–	Jury trial held. Day One (Empanelment) opening statements given; adjourned at 1:15pm until 7/6/94; 9:00AM . . . (sad) [Entry date 08/11/94] [1:88cv1814]
7/6/94	–	Jury trial held. Day Two: hearing held RE:submission of Special Master's report to jury. adjourned until 7/7/94; 9:00AM. . . . (sad) [Entry date 08/11/94] [1:88cv1814]
7/7/94	–	Jury trial held. Day Three; lobby conference. adjourned until 7/8/94; at 9:00Am. (sad) [Entry date 08/11/94] [1:88cv1814]
7/8/94	–	Jury trial held. Day Four . . . cross exam continued redirect, no re cross. adjourned to 7/11/94 9:00Am. . . . (sad) [Entry date 08/11/94] [1:88cv1814]

7/11/94	242	Judge Patti B. Saris. Order entered granting in part, denying in part [122-1] motion for summary judgment of non-infring of the stoll and carroll patents by the one seal SMC Rodless Cylinder granting as to the Carroll patent and denying as to the Stoll patent, granting [187-1] motion for summary judgment of validity and a directed verdict of the reexamined Carroll Patent, granting [191-1] motion for summary judgment and or a directed verdict of infringement of the reexamined Carroll Patent, denying [104-1] motion for summary judgment mtn for stoll patent invalidity, denying [117-1] motion for summary judgment of non-infringement of the stoll patent. cc/cl (scj) [Entry date 07/13/94] [1:88cv1814]
7/11/94	–	Jury trial held. Day Five; adjourned to 7/12/94 at 9:00AM. . . . (sad) [Entry date 08/11/94] [1:88cv1814]
7/12/94	–	Jury trial held. Day Six; defendants move for directed verdict; arguments on defendant's motion for directed verdict/judgment as a matter of law. Motion Denied for reasons stated in court. adjourned until 9:00AM on 7/13/94. . . . (sad) [Entry date 08/11/94] [1:88cv1814]
7/13/94	243	SMC's objections to propsed special verdict form (scj) [Entry date 07/14/94] [1:88cv1814]
7/13/94	244	SMC's objections to proposed jury instructions (scj) [Entry date 07/14/94] [1:88cv1814]

7/13/94	–	Jury trial held. Day Seven. . . . plaintiff's rebuttal case..adjourned to 7/14/94 at 9:00Am . . . (sad) [Entry date 08/11/94] [1:88cv1814]
7/14/94	245	SMC's Objections to Revised Jury Instruction (scj) [Entry date 07/18/94] [1:88cv1814]
7/14/94	246	Special Verdict Form as answered by the Jury (scj) [Entry date 07/18/94] [1:88cv1814]
7/14/94	–	Jury trial held. Day Eight . . . Evidence closed . . . closing arguments; jury charge. . . . 4:00PM verdict returned; motions for directed verdict; both motions Denied. . . . (sad) [Entry date 08/11/94] [1:88cv1814]
7/15/94	–	Jury trial held. Day nine. . . . Conference with counsel re-damages, no chance of settlement. defendant will move for mistrial based on brief deliberations . . . (sad) [Entry date 08/11/94] [1:88cv1814]
7/18/94	–	Jury trial held. Day 10 . . . re; damages. . . . (sad) [Entry date 08/11/94] [1:88cv1814]
7/18/94	–	Status conference held. plaintiff advises it will not pursue willful infringement issue. defendant moves for mistrial based on evidence and short deliberations on liability phase. DENIED . . . Juror injured in slip and fall on [stairs]. recessed until 10:20am . . . opening statements on damages phase. adjourned until 7/19/94. . . . (sad) [Entry date 08/25/94] [1:88cv1814]

7/18/94	252	Judge Patti B. Saris. Clerk's Notes: re: status conference Court Reporter: Cloonan (sad) [Entry date 08/25/94] [1:88cv1814]
7/19/94	–	Jury trial held. Day 11. . . . defendant opening statement. . . . court adjourned at 1:00PM jury excused. . . . (sad) [Entry date 08/11/94] [1:88cv1814]
7/20/94	248	SMC objections to proposed Speecial verdict form and damage jury instructions filed (sad) [Entry date 08/02/94] [1:88cv1814]
7/20/94	–	Jury trial held. Day 11 . . . jury excused; charge conference. . . . (sad) [Entry date 08/11/94] [1:88cv1814]
7/21/94	–	Jury trial held. Day twelve; defendants' closing arguments . . . plaintiffs' closing argument..charged. . . . delebration resume . . . questions to be filed. . . . 2:40pm verdict reached and returned . . . jury dismissed. . . . plaintiff's to file motion for permant injunction. proposed judgment due by 8/19/94 . . . (sad) [Entry date 08/11/94] [1:88cv1814]
7/26/94	247	Judge Patti B. Saris. Post-Trial Memorandum, ENTERED, cc/cl. (ktb) [Entry date 07/27/94] [1:88cv1814]
9/8/94	253	Motion by Smc Pneumatics, Inc. for judgment on the pleadings with respect to the Stoll Patent Non-Infrigement ("sleeve made of a magnetizable material"), filed. (scj) [Entry date 09/09/94] [1:88cv1814]

9/8/94	254	Motion by Smc Pneumatics, Inc. for judgment on the pleadings with respect to the Stoll Patent Non-Infringement ("Two seal-one seal"), filed. (scj) [Entry date 09/09/94] [1:88cv1814]
9/8/94	255	Motion by Smc Pneumatics, Inc. for judgment on the pleadings with respect to the Stoll Patent Invalidity ("Best mode and obviousness") and Carroll Patent Invalidity (Obviousness), filed. (scj) [Entry date 09/09/94] [1:88cv1814]
9/8/94	256	Motion by Smc Pneumatics, Inc. to alter judgment on th[e] grounds that there was no substantial evidence to support the jury's 100% lost profit damages of $4,739,183.00, filed. (scj) [Entry date 09/09/94] [1:88cv1814]
9/8/94	257	Motion by Smc Pneumatics, Inc. for new trial under the Alternative Rule 59, filed. (scj) [Entry date 09/09/94] [1:88cv1814]
9/8/94	258	SMC's Brief concerning a stay pending appeal of any injunction. filed (scj) [Entry date 09/09/94] [1:88cv1814]
9/8/94	259	Response by Smc Pneumatics, Inc. in opposition to prejudgement interest in the amount of $416,727.00, filed. (scj) [Entry date 09/09/94] [1:88cv1814]
9/16/94	263	Judge Patti B. Saris. Notice of Hearing/ conference: set for October 24, 1994 at 3:00. (scj) [1:88cv1814]
9/30/94	–	Terminated document mooting [113-1] motion in limine to exclude testimony and evidence of dfts SMC Copr & SMC

Pneumatice & cntrclm pltf SMC Corpo mooting [115-1] motion to ORDER admitted pltfs admission request no 1-17 denying [144-1] motion to strike [122-1] motion for summery judgment of non-infring of the stoll and carroll patents by the one seal SMC Rodless Cylinder, [117-1] motion for summary judgment of non-infringement of the stoll patent, [104-1] motion for summary judgment mtn for stoll patent invalidity mooting [148-1] motion to stay all mtns and rsps pending denying [185-1] motion to vacate or modify those portions of the Report of the SM in accord w/pltf objection filed and to confirm the remaining portions of the report mooting [212-1] motion for consideration of Special Master's recommendation mooting [229-1] motion in limine to exclude testimony and evidence of Defendants SMC Corporation and SMC Pneumatics, Inc. These motions are moot; see trial transcript and opinion dated 07/26/94. Requested by dw. (dlw) [Entry date 10/17/94] [1:88cv1814]

10/13/94 266 Memorandum by Festo Corp. in opposition to [255-1] motion for judgment on the pleadings with respect to the Stoll Patent Invalidity ("Best mode and obviousness") and Carroll Patent Invalidity (Obviousness), filed. (scj) [Entry date 10/14/94] [1:88cv1814]

10/14/94 272 Memorandum by Festo Corp. in opposition to [253-1] motion for judgment on

the pleadings with respect to the Stoll Patent Non-Infringement ("sleeve made of a magnetizable material"), filed. (scj) [Entry date 10/31/94] [1:88cv1814]

10/14/94 273 Memorandum by Festo Corp. in opposition to [257-1] motion for new trial under the Alternative Rule 59, filed. (scj) [Entry date 10/31/94] [1:88cv1814]

10/14/94 274 Memorandum by Festo Corp. in opposition to SMC's Brief in Support of its objection to prejudgmenet interest, filed. (scj) [Entry date 10/31/94] [1:88cv1814]

10/14/94 275 Memorandum by Festo Corp. in opposition to [257-1] motion for new trial under the Alternative Rule 59, filed. (scj) [Entry date 10/31/94] [1:88cv1814]

10/14/94 276 Memorandum by Festo Corp. in opposition to [258-1] remark, SMC's Brief concerning a stay pending appeal of any injunction filed. (scj) [Entry date 10/31/94] [1:88cv1814]

10/14/94 277 Memorandum by Festo Corp. in opposition to [254-1] motion for judgment on the pleadings with respect to the Stoll Patent Non-Infringement ("Two seal-one seal"), filed. (scj) [Entry date 10/31/94] [1:88cv1814]

10/20/94 267 Reply by Smc Pneumatics, Inc. to response to [253-1] motion for judgment on the pleadings with respect to the Stoll Patent Non-Infringement ("sleeve made of a magnetizable material"), filed. (scj) [Entry date 10/24/94] [1:88cv1814]

10/21/94 268 Reply by Smc Pneumatics, Inc. to response to [257-1] motion for new trial under the Alternative Rule 59, filed. (scj) [Entry date 10/24/94] [1:18cv1814]

10/21/94 269 Reply by Smc Pneumatics, Inc. to response to [255-1] motion for judgment on the pleadings with respect to the Stoll Patent Invalidity ("Best mode and obviousness") and Carroll Patent Invalidity (Obviousness), filed. (scj) [Entry date 10/24/94] [1:88cv1814]

10/21/94 270 Reply by Smc Pneumatics, Inc. to Festo's response to SMC's Brief in support of its Objection to prejudgement interest, filed. (scj) [Entry date 10/24/94] [1:88cv1814]

10/21/94 271 Reply by Smc Pneumatics, Inc. to Festo's response to SMC's Brief concerning a stay pending appeal of any injunction, filed. (scj) [Entry date 10/24/94] [1:88cv1814]

10/21/94 280 Reply by Smc Pneumatics, Inc. to response to [254-1] motion for judgment on the pleadings with respect to the Stoll Patent Non-Infringement ("Two seal-one seal"), filed. (scj) [Entry date 10/31/94] [1:88cv1814]

10/21/94 281 Reply by Smc Pneumatics, Inc. to response to [253-1] motion for judgment on the pleadings with respect to the Stoll Patent Non-Infrigement ("sleeve made of a magnetizable material"), filed. (scj) [Entry date 10/31/94] [1:88cv1814]

10/21/94 282 Memorandum by Smc Pneumatics, Inc. in opposition to Festo's Opposition to SMC's Brief in Support of it's Mo[ti]on to Prejudgement Interest, filed. (scj) [Entry date 10/31/94] [1:88cv1814]

10/21/94 283 Reply by Smc Pneumatics, Inc. to response to [255-1] motion for judgment on the pleadings with respect to the Stoll Patent Invalidity ("Best mode and obviousness") and Carroll Patent Invalidity (Obviousness), filed. (scj) [Entry date 10/31/94] [1:88cv1814]

10/21/94 284 Response by Smc Pneumatics, Inc. in opposition to [274-1] opposition memorandum, filed. (scj) [Entry date 10/31/94] [1:88cv1814]

10/21/94 285 Response by Smc Pneumatics, Inc. in opposition to [273-1] opposition memorandum, filed. (scj) [Entry date 10/31/94] [1:88c1814]

10/21/94 286 Reply by Smc Pneumatics, Inc. to Festo's response to SMC's Rule 50 and Rule 59 Motions concerning Festo's Asserted Lost Profits Damages, filed. (scj) [Entry date 10/31/94] [1:88cv1814]

10/24/94 – Motion hearing held-#296, #297-300, #301 and #257 (scj) [Entry date 11/13/94] [1:88cv1814]

10/24/94 302 Judge Patti B. Saris. Clerk's Notes: re: Post Trial Motions granting [296-1] motion for leave to file response to opposition, denying [297-1] motion for judgment on the pleadings on the Stoll Patent Non-Infringement (Sleeve made of a magnetizable material, denying

[298-1] motion for judgment on the pleadings on the Stoll Patent Non-Infringement (Two Seal-One Seal), denying [299-1] motion for judgment on the pleadings on the Stoll Patent Invalidity (Best Mode and Obviousness and Carroll Patent Invalidity (Obviousness), denying [300-1] motion for judgment on the pleadings concerning Festo's asserted lost profit damages, denying [301-1] motion for new trial, denying [257-1] motion for new trial under the Alternative Rule 59-Fees date to stop as of last day of trial (7/21/94) stay pending appeal w/b denied once injunction enters. Order of injunctions w/b stayed for 45 days. Any supplemental filings due 10/26/94 Court Reporter: Marie Cloonan (scj) [Entry date 11/13/94] [1:88cv1814]

10/26/94 – Judge Patti B. Saris. Endorsed Order entered denying [255-1] motion for judgment on the pleadings with respect to the Stoll Patent Invalidity ("Best mode and obviousness") and Carroll Patent Invalidity (Obviousness). (scj) [Entry date 10/27/94] [1:88cv1814]

10/26/94 – Judge Patti B. Saris. Endorsed Order entered denying [254-1] motion for judgment on the pleadings with respect to the Stoll Patent Non-Infringement ("Two seal-one seal"). (scj) [Entry date 10/27/94] [1:88cv1814]

10/26/94 – Judge Patti B. Saris. Endorsed Order entered denying [253-1] motion for judgment on the pleadings with respect

to the Stoll Patent Non-Infri[n]gement ("sleeve made of a magnetizable material"). (scj) [Entry date 10/27/94] [1:88cv1814]

10/26/94 – Judge Patti B. Saris. Endorsed Order entered denying [256-1] motion to alter judgment on th grounds that there was no substantial evidence to support the jury's 100% lost profit damages of $4,739,183.00. (scj) [Entry date 10/27/94] [1:88cv1814]

10/26/94 – Judge Patti B. Saris. Endorsed Order entered denying [255-1] motion for judgment on the pleadings with respect to the Stoll Patent Invalidity ("Best mode and obviousness") and Carroll Patent Invalidity (Obviousness). (scj) [Entry date 10/31/94] [1:88cv1814]

10/27/94 – Judge Patti B. Saris, by the Court, by Deputy clerk Whitney. Endorsed Order entered granting [249-1] motion for entry of judgment. (SEE SEPARATE JUDGEMENT ENTERED) (scj) [Entry date 10/31/94] [1:88cv1814]

10/27/94 290 Judge Patti B. Saris. Judgment entered for Smc Pneumatics, Inc. against Festo Corp. Principal: $4,322,456.00 , and Prejudgment interest in the amount of $416,727.00. Each party is to bear its own costs and attorneys' fees. Judgmement is hereby entered in favor of Platiff Festo Corporation against Defendants SMC Corporation and SMC Pneumatics, Inc. and each of them on the issues of liability and damages. (scj) [Entry date 10/31/94] [1:88cv1814]

10/28/94	291	SMC's Objections to Proposed Judgement (scj) [Entry date 11/01/94] [1:88cv1814]
10/28/94	292	Response by Festo Corp. in opposition to proposed judgement. (attachment-Irrevocable Standby Letter of Credit Account Form of the Agahi Bank, Limited), filed. (scj) [Entry date 11/01/94] [1:88cv1814]
10/31/94	278	Memorandum by Festo Corp. in opposition to [256-1] motion to alter judgment on th grounds that there was no substantial evidence to support the jury's 100% lost profit damages of $4,739,183.00, filed. (scj) [1:88cv1814]
10/31/94	288	Motion by Smc Pneumatics, Inc. to amend the judgement enterd by the Court on October 28, 1994, filed. (scj) [1:88cv1814]
10/31/94	294	Motion by Smc Pneumatics, Inc. to amend [290-1] judgment order entered on October 28, 1994, filed. (scj) [Entry date 11/03/94] [1:88cv1814]
11/3/94	295	Response by Festo Corp. in opposition to [288-1] motion to amend the judgement enterd by the Court on October 28, 1994, filed. (scj) [Entry date 11/04/94] [1:88cv1814]
11/3/94	304	Response by Smc Pneumatics, Inc. in opposition to [275-1] opposition memorandum filed by Festo concerning SMC's Motion for Relief under Rule 59(e), filed. (scj) [Entry date 11/21/94] [1:88cv1814]

11/9/94	297	Refiling (as suggested by the Court at the October 24, 1994 hearing) of the Motion by Smc Pneumatics, Inc. for judgment on the pleadings on the Stoll Patent Non-Infringment (Sleeve made of a magnetizable material, filed. (scj) [Entry date 11/10/94] [1:88cv1814]
11/9/94	298	Refiling (as suggested by the Court at the October 24, 1994 hearing) of the Rule 50 (b) Motion by Smc Pneumatics, Inc. for judgment on the pleadings on the Stoll Patent Non-Infringement (Two Seal-One Seal), filed. (scj) [Entry date 11/10/94] [1:88cv1814]
11/9/94	299	Refiling (at the suggestion of the Court at the October 24, 1994 hearing) of the Rule 50(b) Motion by Smc Pneumatics, Inc. for judgment on the pleadings on the Stoll Patent Invalidity (Best Mode and Obviousness and Carroll Patent Invalidity (Obviousness), filed. (scj) [Entry date =11/10/94] [1:88cv1814]
11/9/94	300	Refiling of the Rule 50 and Rule 59 Motion (at the suggestion of the Court at the October 24, 1994 hearing) by Smc Pneumatics, Inc. for judgment on the pleadings concerning Festo's asserted lost profits damages, filed. (scj) [Entry date 11/10/94] [1:88cv1814]
11/9/94	301	Refiling of Rule 59 Motion by Smc Pneumatics, Inc. for new trial, filed. (scj) [Entry date 11/10/94] [1:88cv1814]
11/10/94	–	Judge Patti B. Saris. Endorsed Order entered denying [294-1] motion to amend [290-1] judgment order entered

		on October 28, 1994. SMC's motion for a stay of the execution upon the judgment and for the substitution of a letter of credit and agreement (exhibit A) instead of a supercedeas bond is allowed (scj) [Entry date 11/14/94] [1:88cv1814]
11/10/94	– 303	Notice of appeal to Federal Circuit by Smc Pneumatics, Inc. filed. Fee Status: pd Fee Amount: $105.00 Receipt #: 264844 Appeal re: Appeal record due on 12/12/94 (mmh) [Entry date 11/16/94] [1:88cv1814]
11/14/94	–	Judge Patti B. Saris, by the Court, by Deputy Clerk Whitney. Endorsed Order entered denying [301-1] motion for new trial, denying [300-1] motion for judgment on the pleadings concerning Festo's asserted lost profits damages, denying [299-1] motion for judgment on the pleadings on the Stoll Patent Invalidity (Best Mode and Obviousness and Carroll Patent Invalidity (Obviousness), denying [298-1] motion for judgment on the pleadings on the Stoll Patent Non-Infringement (Two Seal-One Seal), denying [297-1] motion for judgment on the pleadings on the Stoll Patent Non-Infringment (Sleeve made of a magnetizable material. (scj) [Entry date 11/16/94] [1:88cv1814]
11/18/94	305	Amended Notice of appeal by Smc Pneumatics, Inc. filed. Fee Status: waived Fee Amount: $105.00: Appeal record due on 12/19/94 (scj) [Entry date 11/23/94] [1:88cv1814]

11/28/94	321	Notice of Docketing Appeal in the Federal Circuit Appeal No. 95-1079 (scj) [Entry date 12/10/94] [1:88cv1814]
12/6/94	306	Transcript of proceedings for 6/29/94, before Judge: Saris. (mmh) [1:88cv1814]
12/6/94	307	Transcript of proceedings for 7/5/94, before Judge: Saris. (mmh) [1:88cv1814]
12/6/94	308	Transcript of proceedings for 7/6/94, before Judge: Saris. (mmh) [1:88cv1814]
12/6/94	309	Transcript of proceedings for 7/7/94, before Judge: Saris. (mmh) [1:88cv1814]
12/6/94	310	Transcript of proceedings for 7/8/94, before Judge: Saris. (mmh) [1:88cv1814]
12/6/94	311	Transcript of proceedings for 7/11/94, before Judge: Saris. (mmh) [1:88cv1814]
12/6/94	312	Transcript of proceedings for 7/12/94, before Judge: Saris. (mmh) [1:88cv1814]
12/6/94	313	Transcript of proceedings for 7/13/94, before Judge: Saris. (mmh) [1:88cv1814]
12/6/94	314	Transcript of proceedings for 7/14/94, before Judge: Saris. (mmh) [1:88cv1814]
12/6/94	315	Transcript of proceedings for 7/18/94, before Judge: Saris. (mmh) [1:88cv1814]
12/6/94	316	Transcript of proceedings for 7/19/94, before Judge: Saris. (mmh) [1:88cv1814]
12/6/94	317	Transcript of proceedings for 7/20/94, before Judge: Saris. (mmh) [1:88cv1814]
12/6/94	318	Transcript of proceedings for 7/21/94, before Judge: Saris. (mmh) [1:88cv1814]
12/6/94	319	Transcript of proceedings for 10/24/94, before Judge: Saris. (mmh) [1:88cv1814]

12/6/94	320	Transcript of proceedings for 11/10/94, before Judge: Saris. (mmh) [1:88cv1814]
12/6/94	–	Order (1) SMC's motion for a stay, pending appeal, is denied. (2) SMC's motion to file a reply is granted. 95-1066 (mmh) [Entry date 12/12/94] [1:88cv1814]
12/8/94	322	Mandate of US Court of Appeals re: [303-1] appeal dismissing [303-1] appeal. Appeal Number 95-1079 (mmh) [Entry date 12/12/94] [1:88cv1814]
1/5/95	–	Case closed. (dlw) [1:88cv1814]
12/14/95	–	Notice of Docketing Record on Appeal from USCA re: received. USCA NUMBER: 95-1066 Case from Federal Circuit (mmh) [Entry date 12/18/95] [1:88cv1814]
12/14/95	324	Judge Patti B. Saris. Judgment entered in the United States Court of Appeals for the Federal Circuit affirming the judgment entered by United States District Judge Patti B. Saris on October 31, 1994. (scj) [Entry date 01/17/96] [1:88cv1814]
4/2/96	326	Judgment entered in the United States Court of Appeals for the Federal Circuit affirming the decision of the United States District Court for the Diestrict of Massachusetts entered on October 31, 1995 from Judge Patti B. Saris. Pre-Judgment interest % (scj) [Entry date 04/16/96] [1:88cv1814]

8/1/96	340	Judge Patti B. Saris . Order entered . . . Accordingly, the total damages due Festo, including post-judgment interest is $7,120,758 in accordance with the Judgment of October 27, 1994, FILED. (cc/cl) (cmg) [Entry date 08/05/96] [1:88cv1814]
8/12/96	–	Judge Patti B. Saris. Endorsed Order entered granting [355-1] motion to stay pending disposition by the Supreme Court. Allowed on the condition that the letter of credit is increased in value to cover th amount FESTO seeks by 9/12/96; issued cc/cl. (sat) [Entry date 08/13/96] [1:88cv1814]
12/22/00	347	Mandate of US Court of Appeals for the Federal Circuit re: reversing appeal 95-1066. (jr) [Edit date 01/03/01] [1:88cv1814]

FEDERAL CIRCUIT

DATE		PROCEEDINGS AND ORDERS
December 12, 1994	8	Received an *amended* notice of appeal (filed on 11/18/94) and certified list from the district court. (bah)
Feb 2, 1995		BRIEF FOR THE APPELLANT, filed. (MS-2/2/95)

Mar 14, 1995	BRIEF FOR THE APPELLEE, filed. (MS-3/14/95) (cr)
Mar 31, 1995	REPLY BRIEF FOR THE APPELLANT, filed. (MS-3/31/95) (cr)
Apr 7, 1995	JOINT APPENDIX, Vols-I-II-III 12 copies each filed, *Vol-III* (Confidential Nonconfidential Vol-III 5 copies filed. (MS-4/7/95) (cr)
April 14, 1995	SUPPLEMENTAL JOINT APPENDIX, Vols-II-&-III 12 copies each field. (MS-4/14/95) (cr) (Per fxg) 6/20/96.
December 14, 1995	AFFIRMED. "JUDGMENT ENTERED" (P). (av) 72 F3d 857
March 6, 1996	APPELLANT - COMBINED PETITION FOR REHEARING AND SUGGESTION FOR REHEARING IN BANC (MS-01/03/96) FILED 01/03/96 PETITION CIRCULATED: 01/18/96 SUGGESTION CIRCULATED: 02/07/98 RESPONSE REQUESTED FROM: APPELLEE (FESTO CORP.) RESPONSE DUE: 02/23/96 RESPONSE FILED: 02/23/96 RESPONSE CIRCULATED: 02/23/96 PETITION/SUGGESTION: DENIED/DECLINED ON 03/06/96. (EOD 03/06/96 BY SCH) 95-1065

April 17, 1997	29.	PETITION FOR WRIT OF CERTIORARI FILED 03/26/96,. SUPREME COURT : 95-1546, CERT. GN ON 03/17/97. 30. JUDGMENT: VAC.RMD ON 03/17/97. U.S.L.W.: 64-3669 (EOD 04/17/97 BY CS) 95-1066
June 9, 1997		APPELLANT - SUGGESTION FOR HEARING IN BANC (MS-04/04/97) FILED: 05/28/97 SUGGESTION CIRCULATED: 05/28/97 SUGGESTION: DECLINED ON 06/09/97. (EOD 06/09/97 BY AV) 95-1066
June 2, 1997	30.	[a] CERTIFIED COPY OF THE U.S. SUPREME CT. JUDGMENT RECEIVED: 06/02/97 (EOD 06/02/97) (AV) 95-1066
June 9, 1997	31	ORDERED: APPELLANTS' SUGGESTION FOR REHEARING IN BANC HAVING BEEN DECLINED, THE CASE IS RETURNED TO THE PANEL. THE MANDATE IS VACATED. SUPPLEMENTAL BRIEFING SCHEDULE SET OUT IN THIS ORDER. BY: CLERK. (CS-06/09/97) FILED: 06/09/97 (EOD 06/09/97) (AV) 95-1066
July 25, 1997		SMC CORP. ET AL. [APPELLANTS] - UPON REMAND FROM THE SUPREME COURT. OPENING BRIEF OF DEFENDANTS-APPELLANTS SMC

	CORP., ET AL. (MS-07/07/97), FILED: 07/07/97. [974282] (EOD 07/25/97 BY SPP) 95-1066
August 12, 1997	FESTO CORPORATION [APPELLEE] – ON REMAND FROM THE UNITED STATES SUPREME COURT, SUPPLEMENTAL BRIEF OF PLAINTIFF-APPELLEE FESTO CORPORATION (MS-08/11/97). FILED: 08/11/97. ~~Rule 33 Notice of Settlement Issued~~. (sp) [974398] (EOD 08/12/97 BY SPP) 95-1066
August 22, 1997	SMC CORP., ET AL. [APPELLANTS] – UPON REMAND FROM THE SUPREME COURT REPLY OF DEFENDANTS-APPELLANTS SMC CORP., ET AL. (MS-08/21/97). FILED: 08/21/97. [974488] (EOD 08/22/97 BY SPP) 95-1066
April 19, 1999	AFFIRMED IN PART, VACATED IN PART, AND REMANDED. PRECEDENTIAL OPINION BY: J. NEWMAN. JUDGMENT ENTERED: 04/19/99. COSTS AGAINST EACH SIDE. (EOD 04/19/99 BY KSH) 95-1066 JUDGMENT VACATED & OPINION OF CT. WITHDRAWN BY CT. ORDER ON 8/20/99.

August 20, 1999	APPELLANT – COMBINED PETITION FOR REHEARING AND SUGGESTION FOR REHEARING IN BANC (MS-05/03/99) FILED: 05/03/99 PETITION CIRCULATED: 05/03/99 SUGGESTION CIRCULATED: 05/13/99 RESPONSE REQUESTED FROM: APPELLEE (FESTO CORP.) RESPONSE DUE: 05/28/99 RESPONSE FILED: 05/28/99 RESPONSE CIRCULATED: 05/28/99 PETITION/SUGGESTION: DENIED/GRANTED ON 08/20/99. (EOD 08/20/99 BY AV) 95-1066
August 20, 1999	APPELLEE – PETITION FOR REHEARING (MS-05/03/99) FILED: 05/03/99 PETITION CIRCULATED: 05/03/99 PETITION: DENIED ON 08/20/99. (EOD 08/20/99 BY AV) 95-1066
October 20, 1999	SMC CORP., ET AL. [APPELLANTS] – IN BANC OPENING BRIEF OF DEFENDANTS-APPELLANTS SMC CORP., ET AL. (MS-10/19/99), FILED: 10/19/99. [001115] (EOD 10/20/99 BY SPP) 95-1066
November 30, 1999 44.	PROCTER & GAMBLE – MOTION OF THE PROCTER & GAMBLE COMPANY FOR AN EXTENSION OF TIME TO FILE AN AMICUS BRIEF. (FS-11/17/99) FILED;

11/17/99. REPLY 1 (46) FILED: 11/22/99. ACTION ON MOTION 47): MOTION GRANTED THE BRIEF IS DUE 12/15/99. BY CT- EN BANC. FILED: 11/30/99, (EOD 11/30/99 BY AV) 95-1066

November 19, 1999 45. PROCTER & GAMBLE CO (AMICUS) – NOTICE THAT APPELLANT'S WILL NOT BE FILING A RESPONSE IN OPPOSITION TO THE MOTION FOR EXTENSION OF TIME TO FILE BRIEF. (FS-11/18/99) RECEIVED: 11/18/99 (EOD 11/19/99) (AMT) 95-1066

December 1, 1999 A.I.P.L.A. [AMICUS] – BRIEF FOR AMICUS CURIAE AMERICAN INTELLECTUAL PROPERTY LAW ASSOCIATION (MS-11/29/99). FILED: 11/29/99. [001162] (EOD 12/01/99 BY AVV) 95-1066

December 2, 1999 FESTO CORPORATION [APPELLEE] – EN BANC RESPONSIVE BRIEF OF PLAINTIFF-APPELLEE FESTO CORPORATION (MS-12/01/99). FILED 12/01/99. ENTRY OF APPEARANCE NOTICE ISSUED. [001163] (EOD 12/02/99 BY SPP) 95-1066

December 2, 1999 48. CERTIFICATE OF INTEREST FOR THE AMICUS CURIAE,

LITTON SYSTEMS INC. (MS-12/01/99) FILED: 12/01/99. (EOD 12/02/99 BY AMT) 95-1066

December 2, 1999 49. HEWLET PACKARD – MOTION OF HEWLETT PACKARD CO FOR LEAVE TO FILE BRIEF AS AMICUS CURIAE IN SUPPORT OF DEFENDANTS-APPELLANTS. (FS-12/01/99) FILED: 12/02/99. ACTION ON MOTION (49); MOOT. LEAVE OF COURT TO FILE AS AMICUS IS NOT REQUIRED IN THIS CASE. BY CLERK. FILED: 12/02/99. (EOD 12/02/99 BY AMT) 95-1066

December 2, 1999 50. CERTIFICATE OF INTEREST FOR THE AMICUS CURIAE, HEWLETT-PACKARD CO. (F-12/01/99) FILED: 12/02/99. (EOD 12/02/99 BY AMT) 95-1066

December 2, 1999 51. CERTIFICATE OF INTEREST FOR THE AMICUS CURIAE, Procter & Gamble Co. (MS-11/30/99) FILED: 12/01/99. (EOD 12/02/99 BY AMT) 95-1066

December 2, 1999 LITTON SYSTEMS, INC. [AMICUS] – EN BANC BRIEF OF AMICUS CURIAE LITTON SYSTEMS, INC. IN SUPPORT OF AFFIRMANCE (MS-12/01/99), filed: 12/01/99,

[001163] (EOD 12/02/99 BY SPP) 95-1066

December 2, 1999 — THE BAR ASSOCIATION OF THE DISTRICT OF COLUMBIA [AMICUS] – AMICUS CURIAE BRIEF OF THE PATENT. TRADEMARK & COPYRIGHT SECTION OF THE BAR ASSOCIATION OF THE DISTRICT OF COLUMBIA IN SUPPORT OF NEITHER PARTY (MS-12/01/99), FILED; 12/01/99. [001164] (EOD 12/02/99 BY SPP) 95-1066

December 2, 1999 — THE PROCTER & GAMBLE COMPANY [AMICUS] – BRIEF FOR AMICUS CURIAE THE PROCTOR & GAMBLE COMPANY SUPPORTING DEFENDANTS-APPELLANTS' POSITION FOR REVERSAL (MS-11/30/99). FILED: 11/30/99, [001165] (EOD 12/02/99 BY SPP) 95-1066

December 3, 1999 — HEWLETT-PACKARD COMPANY [AMICUS] – BRIEF OF HEWLETT-PACKARD COMPANY AS AMICUS CURIAE IN SUPPORT OF DEFENDANTS-APPELLANTS (MS-12/01/99). FILED: 12/01/99, [001166] (EOD 12/03/99 BY SPP) 95-1066

December 3, 1999 — IBM, EATMAN KODAK COMPANY AND FORD MOTOR COMPANY [AMICUS] – BRIEF

FOR AMICI CURIAE INTERNATIONAL BUSINESS MACHINES CORPORATION, EASTMAN KODAK COMPANY AND FORD MOTOR COMPANY (MS-12/01/99). FILED: 12/01/99. [001166] (EOD 12/03/99 BY SPP) 95-1066

December 6, 1999 — THE HOUSTON INTELLECTUAL PROPERTY LAW ASSOCIATION [AMICUS] – BRIEF FOR AMICUS CURIAE OF THE HOUSTON INTELLECTUAL PROPERTY LAW ASSOCIATION REQUESTING AFFIRMANCE IN SUPPORT OF APPELLEE FESTO CORPORATION (MS-12/01/99). RECEIVED: 12/02/99, REJECTED: 12/06/99, CORRECTED BRIEF DUE: 12/20/99, [001166] (EOD 12/06/99 BY SPP) 95-1066

December 9, 1999 — THE HOUSTON INTELLECTUAL PROPERTY LAW ASSOCIATION [AMICUS] – CORRECTED BRIEF FOR AMICUS CURIAE OF THE HOUSTON INTELLECTUAL PROPERTY LAW ASSOCIATION REQUESTING AFFIRMANCE IN SUPPORT OF APPELLEE FESTO CORPORATION (MS-12/01/99), CORRECTED BRIEF RECEIVED:

	12/08/99, FILED: 12/01/99. [001166] (EOD 12/09/99 BY SPP) 95-1066
December 28, 1999	SMC CORP., ET AL. [APPELLANTS] – IN BANC REPLY BRIEF OF DEFENDANTS-APPELLANTS SMC CORP., ET AL. (MS-12/20/99). FILED: 12/20/99. [001186] (EOD 12/28/99 BY SPP) 95-1066
December 28, 1999	SMC CORP., ET AL. [APPELLANTS] – PURSUANT TO LEAVE OF COURT, RESPONSE OF DEFENDANTS-APPELLANTS SMC CORP., ET AL. TO BRIEFS SUBMITTED BY AMICUS CURIAE (MS-12/20/99). FILED: 12/20/99. [001186] (EOD 12/28/99 BY SPP) 95-1066
December 28, 1999	FESTO CORPORATION [APPELLEE] – COMBINED REPLY OF PLAINTIFF-APPELLEE FESTO CORPORATION TO THE BRIEFS OF AMICUS CURIAE (MS-12/20/99). FILED: 12/20/99. [001186] (EOD 12/28/99 BY SPP) 95-1066
January 6, 2000	SMC CORP., ET AL. [APPELLANTS] – UPON REMAND FROM THE SUPREME COURT, JOINT SUPPLEMENTAL APPENDIX (MS-08/27/97). FILED: 12/30/99. [001193]

	(EOD 01/06/00 BY SPP) 95-1066
January 6, 2000	SMC CORP., ET AL. [APPELLANTS] – JOINT APPENDIX (3 VOLUME) (Volume III has confidential and nonconfidential versions) (MS-12/30/99). FILED: 12/~~30~~ (sp) 29/99. [001193] (EOD 01/06/00 BY SPP) 95-1066
November 29, 2000	*Reversed*. Precedential Opinion by J. Schall. in which Mayer, Chief Judge, and Plager, Lourie, Clevenger, Bryson, Gajarsa, and Dyk, Circuit Judges, join; in which Michel, Rader and Linn, Circuit Judges, join with respect TO Parts III-A, III-B, and III-D; and in which Michel, Circuit Judge joins with respect to Part III-E. Concurring opinion by Plager, Circuit Judge; concurring opinion by Lourie, Circuit Judge; Opinion concurring-part and dissenting-in-part with respect to Part III-C by Michel, Circuit Judge, in which Rader, Circuit Judge joins; Opinion concurring-in-part and dissenting-in-part with respect TO Part III-C by Rader, Circuit Judge, in which Michel and Linn, Circuit Judges join; opinion concurring-in-part and dissenting-in-part with respect to Part III-C by Linn, Circuit Judge, in which

Rader, Circuit Judge joins; opinion concurring-in-part and dissenting-in-part with respect to Parts III-A, III-B, III-C, and IV by Newman, Circuit Judge.

Judgment entered: 11/29/00. Costs against each side. (eod by ksh on 11/29/00) 95-1066

February 23, 2001 — 65. THE APPLICATION FOR EXT. OF TIME TO FILE A WRIT OF CERT HAS BEEN PRESENTED TO THE CHIEF JUSTICE, WHO ON 2/13/01, EXTENDED THE TIME TO AND INCLUDING MARCH 29, 2001. (cs-02/13/01) RECEIVED: 02/23/01 (EOD 02/23/01) (BG) 95-1066

March 22, 2001 — 66. THE APPLICATION FOR AN EXTENSION OF TIME WITHIN WHICH TO FILE A CERT PETITION HAS BEEN PRESENTED TO THE CHIEF JUSTICE, WHO ON 3/20/01, EXTENDED THE TIME TO AND INCLUDING 4/9/01. (CS-03/20/01) RECEIVED; 03/20/81 (EOD 03/22/01) (BG) 95-1066

May 8, 2001 — PETITION FOR WRIT OF CERTIORARI FILED 04/09/01, SUPREME COURT #: 00-1543. (EOD 05/08/01 BY BG) 95-1066

June 19, 2001 — PETITION FOR WRIT OF CERTIORARI FILED 04/09/01, SUPREME COURT #: 00-1543, CERT. GRANTED ON

06/18/01. (EOD 06/19/01 BY BG) 95-1066

UNITED STATES DISTRICT COURT
DISTRICT OF MASSACHUSETTS

FESTO CORPORATION, Plaintiff and Counterclaim Defendant, v. SHOKETSU KINZOKU KOGYO KABUSHIKI CO LTD., a/k/a SMC CORPORATION and SMC PNEUMATICS, INC. Defendants and Counterclaim Plaintiffs.	Civil Action No. 88-1814-MA

REPORT OF SPECIAL MASTER
HERBERT F. SCHWARTZ, ESQ.
PURSUANT TO RULE 53(e)(1), F.R.Civ.P.

* * *

b. The Magnetizable Sleeve

Festo asserts that the SMC driven member's non-magnetizable sleeve is an equivalent of the magnetizable sleeve in claim 1 of the Stoll patent. (Festo's Post Trial Brief, pp. 3-7.) In support of this contention Festo urges that (1) the magnetizable sleeve limitation did not appear in the German Stoll patent and should not have been added to the U.S. patent (Festo's Post Trial Brief, p. 5; Tr. 457); and (2) the sleeve material does not affect the operation of the invention. (Festo's Post Trial Brief, pp.

5-6; Tr. 642-645.) Neither of these arguments is convincing.

First, the specification and prosecution history do not support Festo's assertion that the magnetizable limitation should not have been in claim 1. In the specification the sleeve is described as "made of a magnetizable material to reduce magnetic leakage fields." Col. 3, lines 61-62. The prosecution history shows that the patent was filed with a claim 1 that did not have the magnetizable limitation; claim 8, which was dependent from claim 1, had the limitation. In response to a rejection based on Section 112 ¶ 1, claim 1 was rewritten as claim 13 and claim 8 was canceled. (Px 14.) Since consideration was given to whether or not the term should appear in the claim after the first Office Action, Festo's assertion that it was not meant to be there is not persuasive.

Second, the specification contradicts Festo's argument that the sleeve material does not affect the operation of the invention. Festo has tried to characterize the invention as dealing only with sealing, wiping and guiding. (Festo's Post Trial Brief p. 3-4; Tr. 412-413.) Nevertheless, the specification not only describes the sleeve as being made from a magnetizable material but also explains the reason for choosing that material:

> If the driven assembly is provided with a sleeve made of a magnetizable material, which encircles the hollow cylindrical assembly of the magnet arrangement, magnetic leakage fields in the vicinity of the driven assembly can be kept to a minimum. In this way, undesirable braking forces, which would otherwise be generated while piston and driven assembly pass

magnetizable machine components, are avoided. (Px 15 col. 2, lines 23-30.)

Festo performed an experiment which was supposed to demonstrate that regardless, of whether the sleeve was made of a magnetizable or a non-magnetizable material, the operation of the rodless cylinder was the same. (Tr. 697-700; Festo's Post-Trial Brief, p. 6.) SMC criticized the test because it was not performed under load. (Tr. 806-808; SMC's Post-Trial Brief, p. 15.) More importantly, the test shed no light on whether or not both sleeves achieved the advantages that Stoll attributed to having a sleeve made of a magnetizable material: i.e. that both magnetizable and non-magnetizable sleeves served to minimize magnetic leakage fields in substantially the same way.

According to the specification of the Stoll patent, a sleeve made of a magnetizable material achieves the benefit of minimizing magnetic leakage fields. Stoll claims a device with a sleeve made of a magnetizable material. The specification is silent concerning the use of a sleeve made of a non-magnetizable material. Moreover, at the time the application was filed, Stoll thought there was an advantage to using a sleeve made of a magnetizable material. (Tr. 461.)

Although Stoll initially filed a claim broad enough to cover the use of a sleeve made of a magnetizable material, he opted not to pursue such a claim after the first Office Action. As the Federal Circuit said in *Unique Concepts, Inc.* v. *Brown*, 939 F.2d 1558, 1562, 19 USPQ2d 1500, 1504 (Fed.Cir. 1991):

> The statute requires that an inventor particularly point out and distinctly claim the subject matter of his invention. 35 U.S.C. § 112 (1988). It would run counter to this statutory provision for an applicant for patent to expressly state throughout his specification and in his claims [a claim limitation] and then be allowed to avoid that claim limitation in a later infringement suit by pointing to one paragraph in his specification stating an alternative that lacks the limitation, and thus interpret the claim contrary to its plain meaning. Such a result would encourage an applicant to escape examination of a more broadly-claimed invention by filing narrow claims and then, after grant, asserting a broader scope of the claims based on a statement in the specification of an alternative never presented in the claims for examination.[22]

The subject matter disclosed but not claimed in a patent application is dedicated to the public. *Miller* v. *Brass Co.*, 104 U.S. 350 (1881). As the Federal Circuit pointed out in *Unique Concepts*, if the applicant intended to claim an alternative embodiment, he had to persuade the examiner to issue the claim. As is apparent from the Stoll patent prosecution history, Stoll submitted one claim without the magnetizable material limitation and another with the limitation. Both claims were rejected and Stoll decided to only pursue a claim directed to a sleeve made of a magnetizable material. (Px 14.)

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[22] See also *International Visual Corporation* v. *Crown Metal Manufacturing Co, Inc.*, No. 92-1262 (Fed.Cir. 1993) (Lourie, J., concurring at p. 5).

The doctrine of equivalents is the exception and not the rule. In this case, the meaning and scope of claim 1 of the Stoll patent was clear. Stoll consciously chose to not claim an embodiment of his invention that used a magnetizable sleeve. Festo cannot now try to use the doctrine of equivalents to recapture a claim that Stoll abandoned during prosecution "because other players in the marketplace are entitled to rely on the record made in the Patent Office in determining the meaning and scope of the patent." *Lemelson* v. *General Mills, Inc.*, 968 F.2d 1202, 1208, 23 USPQ2d 1284, 1289 (Fed.Cir. 1992).

Festo's commercial embodiment, like SMC's, uses a non-magnetizable sleeve. (Px 118, 210.) However, the patent claim, and not any commercial embodiment, serves as notice to others of the patentees rights. SMC had a right to rely on the claim language, in light of the specification and the prosecution history, and to design around it. *London* v. *Carson Piri Scott & Co.*, 946 F.2d 1534, 20 USPQ2d 1456 (Fed.Cir. 1991), *supra*, accord, *State Industries* v. *A.O. Smith Corp.*, 751 F.2d 1226, 224 USPQ 418 (Fed.Cir. 1985). Therefore, claims 1-3 and 5-7 of the Stoll patent are not infringed by the SMC rodless cylinders.

IV. DAMAGES

A. FESTO'S MOTION TO INCLUDE DAMAGES PROOF

The majority of SMC's sales were made in the years 1989-1991. (Tr. 1388; Dx 494.) This information was produced on August 31, 1992 pursuant to an Order of the Special Master. (Dkt. 172.) To protect the confidentiality

of SMC's sales data, Festo used an independent accountant to testify about damages at the hearing on damage.

There was a dispute between the parties as to whether Festo should be permitted to introduce into evidence the testimony by its accountant, Steven Skalak, and related exhibits taken *de bene* at the hearing on damages because basis for the new evidence was not included in the pretrial order.

* * *

> sales denied to him is the measure of his loss, and the Infringer is liable for that. *Lam, Inc.* v. *Johns-Manville Corp.*, 718 F.2d 1056, 1065, 219 USPQ 670, 675 (Fed.Cir. 1983) quoting *Livesay Window Company* v. *Livesay Industries*, 251 F.2d 469, 471-72, 116 USPQ 167, 169 (5 Cir. 1958).

Festo and SMC were the only suppliers of sealed magnetically coupled rodless cylinders during the period of infringement. (Tr. 32.) There was a demand for the magnetically coupled rodless cylinder because of their unique characteristics. (Tr. 28-30.)

Mechanical rodless cylinders, like the band and cable cylinders, did not offer the advantages of the magnetically coupled rodless cylinder. (Tr. 1415-1425, 1844-1860.) Among these advantages are the ability to be leak proof and the ability to work without special lubricated operating fluid. Moreover, Festo could have met the demand for increased rodless cylinders. (Tr. 1427, 1431.)

Festo failed to prove, however, that it lost any profits. For seven out of ten years reported Festo had operating

losses. (Tr. 1643, 1755-1756.) For every dollar of Festo sales, forty to fifty cents represents the cost of the product and fifty to sixty cents represents the selling expenses. (Tr. 1916-1918; Dx 102.) Festo did not show that the DGOs were any more profitable than Festo's other products. (Tr. 1638-1639.) Thus Festo did not demonstrate that, on a straight dollar average, any increase in sales would have led to an increase in profits. (Tr. 1640, 1691-1692.) Festo did not offer proof to support the argument that on an incremental basis additional sales would have created profits.

Festo asserted that a company does not have to be profitable to recover lost profits. Festo relied on *Saf-Gard Products, Inc.* v. *Service Parts, Inc.*, 491 F.Supp. 996, 206 USPQ 976 (D.Ariz. 1980) for this proposition. This reliance is misplaced. In *Saf-Gard*, the plaintiff was awarded lost profits based on a price erosion theory. The infringing defendant's presence and actions in the marketplace drove down the price of the product, making the plaintiff less profitable and, ultimately, bankrupt. The court awarded damages based on the uneroded price. 491 F.Supp. at 1009, 206 USPQ at 986-987. Festo did not claim price erosion. In fact, Festo's rodless cylinders remained more expensive than SMC's throughout the infringement period. (Tr. 1437, 1524.)

Festo has simply failed to show that it would have netted anything had it made SMC's sales. There is no basis for awarding damages under a lost profits theory.

2. Reasonable Royalty

When a patent owner cannot establish lost profits or an established royalty, it is entitled to a reasonable royalty. This is characterized as the amount that would have been set at the time the infringement began between a hypothetical willing patent owner and a willing potential

* * *

Lantech, Inc., 807 F.2d 970, 1 USPQ2d 1202 (Fed.Cir. 1986). When the patentee makes substantive changes, the reexamined claims are treated like reissue claims under 35 U.S.C. § 252. "Thus a person practicing a patented invention would not be considered an infringer for the period between issuance of an invalid patent and the conversion through reexamination to a valid patent." *Fortel Corp.* v. *Phone-Mate, Inc.*, 825 F.2d 1577, 1579, 3 USPQ2d 1771, 1773 (Fed.Cir. 1987), quoting H.R. Rep. No. 1307, 96th Cong., 2d Sess. 8 (1980).

During reexamination of the Carroll patent, Festo canceled the only independent claim and added a new independent claim. See *supra*. pp. 15-17. This was a substantial change. As a result, the damages are based on SMC sales made from the time the reexamination certificate issued (October 25, 1988) to the expiration of the reexamined Carroll patent (December 18, 1990). The total infringing SMC sales was $1,470,127. A fifteen percent royalty equals $220,519.

1. Prejudgment Interest

35 U.S.C. § 284 authorizes the award of "interest . . . as fixed by the court." In *General Motors Corp.* v. *Devex Corp.*, 461 U.S. 648, 657, 217 USPQ 1185, 1189 (1983), the Supreme Court found that awarding prejudgment interest was within the discretion of the trial court, but that prejudgment interest should ordinarily be awarded

* * *

Thus, Festo's contention is unfounded. SMC sought the advice of outside counsel. SMC's counsel reviewed the patents and their prosecution histories and formed a competent legal opinion regarding infringement of the Stoll patent and validity of the Carroll patent. Festo has not proved willful infringement by clear and convincing evidence by SMC.

A. ATTORNEY'S FEES

35 U.S.C. § 285 states. "The court in exceptional cases may award reasonable attorney's fees to the prevailing party." Festo has asked for an award of attorney's fees based on its contention that SMC's infringement was willful. (Fest Post Trial Brief, p. 48.) SMC has also requested an award of attorney's fees based on its contention that several of Festo's arguments were frivolous and that Festo did not disclose the best mode of making the Stoll patent. (SMC's Post Trial Brief, pp. 47-48.)

This lawsuit was hotly contested. However, this is not an exceptional case meriting the award of attorney's

fees. I recommend that each side bear its own costs and attorney's fees.

VI. SUMMARY JUDGMENT AND DIRECTED VERDICT MOTIONS

SMC moved for summary judgment of invalidity of the Stoll patent based on failure to disclose the best mode. (Dkt. 104.) SMC moved for summary judgment of noninfringement of the Stoll patent based on its use of a non-magnetizable sleeve. (Dkt. 117.) SMC also moved for summary judgment of noninfringement of the reexamined Carroll patent and the Stoll patent based on its use of one seal, instead of two. (Dkt. 122.) At the conclusion of plaintiff's evidence, SMC moved for a directed verdict. (Tr. 866.) SMC renewed this motion at the conclusion of the hearing on liability (Tr. 1302.)

It is proper to grant summary judgment under Rule 56(c), F.R.Civ.P. if " the pleadings, depositions, answers to interrogatories, and admissions on file, together with the affidavits, if any, show that there is no genuine issue as to any material fact and that the moving party is entitled to a judgment as a matter of law." The Supreme Court has said that under this standard, "the mere existence of *some* alleged factual dispute between the parties will not defeat an otherwise properly supported motion for summary judgment; the requirement is that there be no *genuine* issue of *material* fact. *Anderson* v. *Liberty Lobby, Inc.*, 477 U.S. 242, 247-48 (1986) (emphasis in original).

The Federal Circuit has found that "[s]ummary judgment is as available in patent cases as in other areas

of litigation." *Continental Can Co. USA, Inc.* v. *Monsanto Co.*, 948 F.2d 1264, 1265, 20 USPQ2d 1746, 1747 (Fed.Cir. 1991). Of course, the court must "view the evidence presented in a light most favorable to the nonmoving party" and "draw all reasonable inferences in favor of the nonmoving party." *C.R. Bard, Inc.* v. *Advanced Cardiovascular Systems*, 911 F.2d 670, 672, 15 USPQ2d 1540, 1542 (Fed.Cir. 1990). "The burden of establishing summary judgment is with the movant, with due consideration to the burdens and presumptions at trial." *Laitram Corp.* v. *NEC Corp.*, 952 F.2d 1357, 1363, 21 USPQ2d 1276, 1280-1281 (Fed.Cir. 1991).

The Supreme Court has adopted the view that the standard for summary judgment mirrors the standard for a directed verdict. *Anderson* at 250. Thus "the trial judge must direct a verdict if, under the governing law, there can be but one reasonable conclusion as to the verdict." *Id.* Directed verdicts are now called judgments as a matter of law under Rule 50, F.R.Civ.P., but this standard has not changed. 1991 Advisory Committee Notes. Because the motion for a directed verdict was made to a Special Master, and not a trial judge, I have also considered this motion under Rule 52(c)[24] as a motion for judgment on partial findings. Rule 52(c) parallels Rule 50(a), allowing the court to enter judgment when it can appropriately make a disposition on the evidence.

In light of the findings of fact and recommendations made in this report after the hearings herein, I

[24] Before December 1, 1991 such a motion was available under Rule 41(b), F.R.Civ.P. See, e.g., *Pitts* v. *United States* 599 F.2d 1103 (1 Cir. 1979).

recommend that the motion concerning invalidity on best mode of the Stoll patent be denied. I also recommend that the notion concerning the noninfringement of both patents based on SMC's use of one seal be denied.

Based on the analysis contained in this report concerning the magnetizable sleeve limitation in the Stoll patent claims, summary judgment and a directed verdict or judgment on partial findings in favor of SMC is appropriate with respect to SMC's noninfringement of claims 1-3 and 5-7 of Stoll.[25] Festo and SMC agree that the SMC driven member has a non-magnetizable sleeve. Therefore there is no dispute as to any material fact. The dispute centers on the meaning of the limitation in the claim.

Claim construction is a question of law for the court. *Loctite Corp.* v. *Ultraseal, Ltd.*, 781 F.2d 861, 228 USPQ 90 (Fed.Cir. 1985). To properly construe the claims, the court should look to the specification, file history, other claims and expert testimony. *Whittaker Corp.* v. *UNR Industries, Inc.*, 911 F.2d 709, 15 USPQ2d 1742 (Fed.Cir. 1990); *Lemelson* v. *General Mills, Inc.*, 968 F.2d 1202, 1208, 23 USPQ2d 1284, 1289 (Fed.Cir. 1992). As demonstrated *supra*. pp. 47-51, when properly construed the Stoll claims cannot be read to cover the non-magnetizable sleeves used in the SMC rodless cylinders. Therefore I recommend that SMC's motion for summary judgment

[25] The discussion in this report makes plain that there is no basis for granting Festo's motion for a directed verdict. (Tr. 1302.) And, in any event, Festo never briefed such a motion as it said it would. (Tr. 1308.)

and its motion for directed verdict of non-infringement of claims 1-3 and 5-7 of Stoll be granted.

VII. CONCLUSION

SMC's rodless cylinders do not infringe any of claims 1-3 or 5-7 of the Stoll patent. The SMC cylinders with cushions, in both the basic and the slider lines, and two seals literally infringe claims 5, 6 and 9 of the reexamined Carroll patent. The SMC cylinders with cushions, in both the basic and the slider lines, and one seal infringe claims 5, 6 and 9 under the doctrine of equivalents. Based on this infringement of the reexamined Carroll patent, I recommend that damages and prejudgment interest in the amount of $251,850 be awarded.

/s/ Herbert F. Schwartz
Herbert F. Schwartz, Esq.
Special Master

April 27, 1993

IN THE UNITED STATES DISTRICT COURT
FOR THE DISTRICT OF MASSACHUSETTS

FESTO CORPORATION, Plaintiff and Counterclaim Defendant, v. SHOKETSU KINZOKU KOGYO KABUSHIKI CO., LTD., a/k/a SMC CORPORATION, and SMC PNEUMATICS, INC., Defendants and Counterclaim Plaintiffs.	Civil Action No. 88-1814-MA BBO #549137

FESTO CORPORATION'S OBJECTIONS TO REPORT OF SPECIAL MASTER HERBERT F. SCHWARTZ, ESQ. PURSUANT TO RULE 53(e)(3), F.R. CIV. P.

* * *

invention includes a number of functionally and structurally cooperatively assembled components of which the sleeve is merely one of many. This is readily apparent from reading claim 1 of the patent. The solution of the wiping, guiding and sealing problems as defined in Claim 1 of the Stoll patent involved the use of separate guide and seal rings, as acknowledged by the Special Master on pages 27 and 38-39 of his Report. The separate rings act in cooperation with the remaining elements of the claim to carry out the functions of the invention in

wiping, guiding and sealing in order to operate in keeping dirt and impurities from interfering with the functioning of the rodless cylinder.

The sleeve on the outer or driven member serves to encircle and assist in holding the rings and magnets in position. Whether it is made of steel (magnetizable) or aluminum (non-magnetizable) does not matter for the recited functions of the claimed invention including wiping, guiding and sealing. (TR 411-413; 642-645).

The gratuitous inclusion of the words relating to the sleeve as being formed "of a magnetizable material" was not necessary and not intended by the inventor to have been included and should not be considered as material to the interpretation of the claim for purposes of infringement. (TR 412). Dr. Stoll should not be held to unintentional limitations added incorrectly by an attorney of which he was not aware and of which he certainly did not intend. In the remainder of the world, Stoll's corresponding patents do not have a sleeve material restriction in their main claim. (TR 412). The claim should clearly be interpreted to hypothetically omit the language "of a magnetizable material". The law clearly permits such an interpretation as discussed below.

Minimizing braking forces due to magnetic field leakage is incidental to the invention and not included in the claim. Dr. Wolf pointed this out in his testimony (TR 820-821). Dr. Stoll also clearly explained that for purposes of the invention, the sleeves are the same. (TR 411-413; 535-539). In fact, the difference in minimizing magnetic field leakage braking forces is insubstantial as pointed out by the uncontroverted testimony of Dr. Wolf. The

video film presented as evidence by Festo through Dr. Wolf displaying two cylinders operating in synchronization for a period of time shows no difference in braking force between the steel and aluminum sleeves in operation. (Px P-22).

It must be emphasized that claim 1 does not require and does not even mention a function of minimizing magnetic field leakage to carry out the invention.

There is no prior art basis for limiting the interpretation of the claim in this manner. There was no file wrapper estoppel. The Examiner presented no prior art to limit this interpretation

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carrying out the invention and then attempted to invalidate the patent on that basis. (TR 971, 988). Accordingly, no prior art was presented and none is known which would require the Stoll invention to include a recitation of the magnetizable material for the purpose of minimizing magnetic leakage fields or any other purpose. A hypothetical Claim 1 eliminating the words "of a magnetizable material" is clearly patentable over the prior art. In rejecting SMC's failed attempt to invalidate Claim 1 of the Stoll patent independent of whether the sleeve is magnetizable or not, the Master clearly pointed out the numerous ways in which the patent claims in issue were valid independent of the sleeve material. (Master's Report pgs. 31-42).

Accordingly, there was no reason for Dr. Stoll to include a gratuitous recitation such as magnetizable sleeve material in his broadest claim, Claim 1. He has

clearly testified that this was not the case. He never intended to do so and it should not have been done. (TR 457). The prior art does not require it and Dr. Stoll did not choose to include it. It was mistakenly included by the attorney prosecuting the application before the United States Patent and Trademark Office without consultation with the inventor. (TR 455). It certainly does not represent the inventor's wishes and he should not be bound by it.

The Examiner at the Patent and Trademark Office certainly did not require that the recitation be included in the claim. He

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UNITED STATES DISTRICT COURT
DISTRICT OF MASSACHUSETTS

FESTO CORPORATION, Plaintiff and Counterclaim Defendant, v. SHOKETSU KINZOKU KOGYO KABUSHIKI CO. LTD., a/k/a SMC CORPORATION and SMC PNEUMATICS, INC. Defendants and Counterclaim Plaintiffs.	Civil Action No. 88-1814-PBS

PLAINTIFF FESTO CORPORATION'S
PRETRIAL MEMORANDUM UNDER L.R. 16.5

* * *

100. Festo has its main headquarters on Long Island, New York and has about 100,000 square feet in its building.

101. Mr. Kurt Stoll is a graduate mechanical engineer and received his diploma in 1960. He is one of the owners of Festo KG, the parent of Plaintiff Festo. He is also a Director of Festo KG. He and his brother, Wilfried Stoll, are Managing Partners of Festo KG.

102. The products sold in the United States are manufactured and sold by Festo Corporation, the U.S. subsidiary of Festo KG.

FACTS WHICH FESTO INTENDS TO PROVE AT TRIAL IN SUPPORT OF EACH CLAIM

1. The Carroll and Stoll patents were duly and legally issued and are valid.

2. The making, assembly and selling of magnetically coupled rodless cylinders in this judicial district and elsewhere in the United States by SMC Pneumatics constitute direct and contributory infringement of, and active inducement to infringe, the Carroll and Stoll patents.

3. The shipping of parts of the magnetically coupled rodless cylinder into the United States for assembly here, and the selling of magnetically coupled rodless cylinders in this judicial district and elsewhere in the United States by SMC Corp.

* * *

and the magnets on the piston so that there would be minimal resistance to movement of the piston within the tube under pneumatic or hydraulic force. Additionally, it was necessary to seal the tube and to wipe dirt particles or contaminants within the tube away from the area where the magnets are located. In this way, the possibility of reducing the magnetic attraction force due to contamination or dirt is minimized.

91. Prior art rodless cylinders incorporated a single ring on the piston which was used to not only maintain the separation between the piston and the inner wall surface of the tube but also to create a seal with the inner wall surface so that fluid pressure can build up within the tube on one side of the piston to move the piston. A more rigid or harder material used for the seal is desirable in order to maintain the separation between the piston and the inner wall surface of the tube. However, when a more rigid or harder material is used for the seal, the sealing and wiping functions are not effectively obtained. On the other hand, if a soft resilient material is used for the seal, it will conform well to the engaging surface of the inner wall of the cylinder tube and seal and wipe effectively; but, the magnetic force and free floating piston will overcome the soft material and the magnets will engage the tube wall and interfere with the reciprocal movement of the piston in the tube.

92. Mr. Stoll conceived of a solution to the above problems. Mr. Stoll divided the task of accomplishing these separate functions into separate means positioned on the piston.

93. Mr. Stoll conceived of the idea of using an inner guide ring on the piston, which ring is made of a more rigid but slidable material, to maintain the proper spacing between the piston magnets and the wall of the tube while the piston reciprocately moves in the tube. Mr. Stoll further conceived of the idea of using a wiper ring adjacent to the guide ring where the wiper ring is made from a soft resilient material, which carries out the primary wiping and sealing function.

94. There is some overlap in function between the wiping and sealing ring and the guide ring and the ultimate result is a very effective magnetically coupled rodless cylinder which will have a long operational life.

95. The above problem and solutions are fully described in U.S. Patent No. 4,354,125, which issued on October 12, 1982 to Kurt Stoll.

96. The claims of the Stoll patent specifically define Mr. Stoll's invention in regard to the guiding, wiping and sealing functions.

* * *

DEFENDANT'S EXHIBIT 105

UNITED STATES DISTRICT COURT
DISTRICT OF MASSACHUSETTS

FESTO CORPORATION, Plaintiff and Counterclaim Defendant, v. SHOKETSU KINZOKU KOGYI KABUSHIKI CO., LTD., a/k/a SMC CORPORATION, and SMC PNEUMATICS, INC., Defendants and Counterclaim Plaintiffs.	Civil Action No. 88-1814-MA BBO #549137

PLAINTIFF FESTO CORPORATION'S
TRIAL MEMORANDUM

* * *

B. SMC's Infringement Of The Carroll Patent.

As will be established at trial, SMC has literally infringed claims 9, 5 and 6 of the Carroll patent. In other words, the accused SMC devices include each and every element recited in the above claims.

C. SMC's Infringement Of The Stoll Patent.

Most of the accused SMC devices include each and every element recited in claims 1-3 and 5-7 of the Stoll

patent with the exception of the composition of the cylindrical sleeve which encircles the tube of the driving member. Specifically, the sleeves of the accused devices are made from a non-magnetizable material, such as an aluminum alloy, instead of a magnetizable material as recited in the claims. In addition, some of the accused devices apparently employ only one sealing ring instead of "rings" as claimed.

1. The substitution of a non-magnetizable sleeve for a magnetizable sleeve does not substantially change the operation of the device as claimed.

The courts have addressed situations on all fours with the case at bar. In *Graver Tank, supra*, which is considered *the* classical equivalents decision, the patentee claimed a fusible composition containing "a major proportion of one or more alkaline earth metal silicates." The

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thereby be well equipped to determine whether SMC has made a change of such substance as to make the doctrine of equivalents inapplicable, or whether they have simply replaced a sleeve of one type of material with one of a different material that does not cause the invention to operate in a materially different manner.

2. The use of one sealing ring, as opposed to more than one ring, for wiping the internal wall of the tube containing the piston is not a change of such substance as to make the doctrine of equivalents inapplicable.

It is well settled that "infringement is not avoided by making into one part that which has been shown as two." *No-Joint Concrete Pipe Co. v. Hanson*, 344 F.2d 13, 15 (9th Cir. 1965). SMC has nevertheless attempted to avoid infringement of the Stoll patent by using only one sealing ring in some of its rodless cylinders rather than more than one. It is clear that the unit will be sealed, thereby allowing the piston to be moved by pneumatic pressure, regardless of the number of sealing rings used. It is also clear that one sealing ring will provide wiping action as described in the Stoll patent. It will accordingly be shown at trial that SMC's units are functionally equivalent to the patented Festo construction.

The use of two seals may provide certain additional advantages over one, such as more efficient wiping. The court addressed the same situation in the *No-*

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UNITED STATES DISTRICT COURT
DISTRICT OF MASSACHUSETTS

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PLAINTIFF FESTO CORPORATION'S
TRIAL BRIEF

* * *

alloy materials. Festo will also provide evidence that the sleeve material of the SMC device performs substantially the same function, in substantially the same way to achieve substantially the same result as the claimed invention. More specifically. the Stoll invention is directed to wiping, guiding and sealing, i.e. wiping dirt and impurities away from the magnets of the piston and driven member, guiding the piston and driven member inside and outside the cylinder tube, respectively, and sealing the piston to prevent "blow-by". Accordingly, the material used for the sleeve which encircles the outer

magnet arrangement is inconsequential to the Stoll invention of wiping, guiding and sealing. Festo will show at trial that the specific arrangement of the guide rings and seals are not affected by the driven member sleeve material.

A function attributable to the sleeve material in the specification of the Stoll patent will be shown by Festo to not be part of the claimed invention; however, even if one were to include the function of keeping magnetic leakage fields in the vicinity of the driven member to a minimum, Festo will prove at trial that the aluminum alloy sleeve material will function in substantially the same way to achieve substantially the same result as a magnetizable sleeve material.

The jury will thereby be well equipped to determine whether SMC has made a change of such substance as

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UNITED STATES DISTRICT COURT
DISTRICT OF MASSACHUSETTS

FESTO CORPORATION, Plaintiff, v. SHOKETSU KINZOKU KOGYO KABUSHIKI CO., LTD., a/k/a SMC CORPORATION and SMC PNEUMATICS, INC., Defendants.	CIVIL ACTION NO. 88-1814-PBS

ORDER RE: PRETRIAL MOTIONS

July 11, 1994

Saris, U.S.D.J.

On February 3, 1994, this Court heard oral argument on the following motions: The motions by SMC Corporation and SMC Pneumatics, Inc. (collectively "SMC") for summary judgment on non-infringement of the Stoll Patent, non-infringement of the Carroll Patent, and invalidity of the Stoll Patent[1]; and the motions by Festo Corporation ("Festo") for summary judgment and for a

[1] Prior to the Master's Hearing, SMC filed motions for summary judgment on non-infringement of the Stoll Patent, non-infringement of the Stoll and Carroll Patents by the one seal SMC rodless cylinder, and invalidity of the Stoll Patent. Following the Master's Hearing, SMC filed a motion for summary judgment and a directed verdict on non-infringement of the Stoll Patent.

directed verdict on infringement of the Carroll patent, validity of the Carroll patent, infringement of the Stoll patent, and validity of the Stoll patent. As discussed in court that day, and as explained in a Procedural Order issued on February 18, 1994, all motions for summary judgment were denied, with the exception of the motions concerning infringement of the Reexamined Carroll patent. After additional briefing on that issue, this Court now orders that Festo's and SMC's motions are each allowed and denied in part. This opinion also addresses the use of the Master's Report ("Report") at trial, and assumes familiarity with that report. The Court also explains the reasons for denial of the motions concerning infringement of the Stoll Patent under the doctrine of equivalents, which SMC urged the court to reconsider.

1. Master's Report

On May 21, 1992, pursuant to Fed. R. Civ. P. 53(e)(3), Judge Mazzone referred this case to Special Master Herbert F. Schwartz, Esq. for the purpose of considering all claims presented by the parties. In the Order of Reference, Judge Mazzone explained that the appointment of a special master was warranted because "[n]ot only are the claims complex, but the history of this litigation, the voluminous record, and the contentions of the parties promise that this case will require an inordinate amount of judicial time and resources at a time when this Court has four pending vacancies and my own schedule is severely taxed."

SMC filed a limited objection to the reference, primarily directed to the contention that the Master's

authority be no broader than that provided under Fed. R. Civ. P. 53(e). Hearings on the issues of infringement, invalidity and enforceability of the reexamined Carroll and Stoll patents were held before the special master on September 14-21 and hearings on damages were held on October 19-22 and on November 9, 1992. The Master issued his report on April 27, 1993. Both sides filed objections in a timely fashion.

Fed. R. Civ. P. 53 (b) provides that a reference to a master shall be "the exception and not the rule." "In actions to be tried by a jury, a reference shall be made only when the issues are complicated. . . . " *Id.* In an action to be tried by a jury, the master's "findings upon the issues submitted to the master are admissible as evidence of the matters found and may be read to the jury, subject to the ruling of the court upon any objections in point of law which may be made to the report." Fed. R. Civ. P. 53(e)(3).

"[T]he court has the power to appoint masters without the consent of the parties." *Constant* v. *Advanced Micro-Devices Inc.*, 848 F.2d 1560, 1566 (Fed. Cir.) ("*Constant*"), *cert. denied*, 488 U.S. 892 (1988). "Reference of issues to a master, even compulsory reference, does not violate the 7th Amendment right to trial by a jury to which a master's findings may be read under Rule 53(e)(3)." *Id.* (citing *Crateo, Inc.* v. *Intermark, Inc.*, 536 F.2d 862, 868 (9th Cir.), *cert. denied*, 429 U.S. 896 (1976)). *See also Ex parte Peterson*, 253 U.S. 300, 312 (1920).

"Where complicated issues of patent law are involved, the appointment of an experienced patent attorney is quite appropriate." *Constant*, 848 F.2d at 1567.

"Masters can properly aid the court in evaluating issues of patent validity and infringement in the context of motions for summary judgment, and have often done so." *Id.* at 1566. *Cf. International Nickel Co.* v. *Ford Motor Co.*, 166 F. Supp. 551, 553 (S.D.N.Y. 1958) (reference to master in highly technical patent case). "Masters have been employed by courts (under Fed. R. Civ. P. 53 (e)(3)) to explain technology to the jury or make findings on specific issues which are then read to the jury." *Structural Rubber Products Co.* v. *Park Rubber Co.*, 749 F.2d 707, 720 (Fed. Cir. 1984).

In jury cases, some courts have held that the master's findings constitute prima facie evidence to be submitted to the jury for its consideration. *Coyner* v. *United States*, 103 F.2d 629, 634-35 (7th Cir. 1939). *See generally* Irving R. Kaufman, *Masters in the Federal Courts: Rule 53*, 58 Columbia L. Rev. 452, 460 (1958).

SMC has filed a motion in limine objecting to submission of the master's fact findings to the jury. This court has determined that the master's report will be submitted to the jury because of the complexity of the issues and technology involving magnetically coupled rodless cylinders and two different patents. Due to the length of the report, the number of issues, and the complexity of the technology, the Court will allow the jury to read the report by themselves rather than have the report read orally to them. However, in order to ensure that the master's report not receive undue weight, it will not remain with the jurors while deliberating. The attorneys have conferred and agreed to sanitize the report to delete contentions of the parties, and issues which the court has ruled on as a matter of law.All disputes were resolved by

the court. Finally, the Court will not instruct the jury that the master's report is "prima facie" evidence, as that language is missing from Rule 53. Rather, the Court will instruct the jury that it may determine what weight, if any, to give the master's findings.

2. Infringement Under Doctrine of Equivalents of Stoll Patent Concerning Magnetizable Sleeve Limitation

Based on the analysis contained in the report concerning the magnetizable sleeve limitation in claim 1 of the Stoll patent, the master recommended that SMC's motion for summary judgment should be granted with respect to SMC's infringement of claims 1-3 and 5-7 of Stoll. SMC has moved that this Court adopt the master's recommendation with respect to non-infringement of the Stoll patent. Festo has moved for summary judgment and/or a directed verdict of infringement of the Stoll patent.

"In order to determine a question of patent infringement, a district court must (1) determine as a matter of law the scope and meaning of the claims at issue and (2) determine as a factual matter whether the properly construed claims encompass or 'read on' the accused device." *Vaupel Textilmaschinen* v. *Meccanica Euro Italia*, 944 F.2d 870, 879 (Fed. Cir. 1991) ("*Vaupel*"). *See also ZMI Corp.* v. *Cardiac Resuscitator Corp.*, 844 F.2d 1576, 1578 (Fed. Cir. 1988). "To ascertain the meaning of claims, [the court considers] three sources: the claims, the specification, and the prosecution history." *Unique Concepts, Inc.* v. *Brown*, 939 F.2d 1558, 1561 (Fed. Cir. 1991) ("*Unique*"). To

constitute an infringement, the accused device must contain each and every limitation of the claim or its substantial equivalent. *Vaupel*, 944 F.2d at 879. "An equivalent must be found for every limitation of the claim somewhere in an accused device, but not necessarily in a corresponding component, although that is generally the case." *Corning Glass Works* v. *Sumitomo Elec. U.S.A., Inc.*, 868 F.2d 1251, 1259 (Fed. Cir. 1989) (adding negative dopant to the cladding is equivalent to adding positive dopant to the core in the context of the claimed invention).

The court agrees with the conclusion of the Master that "when properly construed the Stoll claims cannot be read to cover the non-magnetizable sleeves used in the SMC rodless cylinders." Report, pp. 47-50, 68. Thus, there is no literal infringement as a matter of law. Festo does not dispute this ruling.

However, the court disagrees that summary judgment can be granted with respect to the claim of infringement by equivalents. A finding of infringement by equivalents requires a determination whether the accused device performs substantially the same function in substantially the same way to achieve substantially the same result. *Unique*, 939 F.2d at 1564. *See generally Graver Tank & Mfg. Co.. Inc.* v. *Linde Air Prods. Co.*, 339 U.S. 605, 608 (1950). "This equitable doctrine evolved from a balancing of competing policies, each of which supports the Constitutional purpose of promoting the 'useful arts.' " *London* v. *Carson Pirie Scott & Co.*, 946 F.2d 1534, 1538 (Fed. Cir. 1991) ("*London*") (citing U.S. Const. art. I, § 8, cl. 8).

> On the one hand claims must be 'particular' and 'distinct,' as required by 35 U.S.C. § 112, so that the public has fair notice of what the patentee and the Patent Trademark office have agreed constitute the metes and bounds of the claimed invention. . . . On the other hand, the patentee should not be deprived of the benefits of his patent by competitors who appropriate the essence of an invention while barely avoiding the literal language of the claims. . . . Thus, where an infringer, instead of inventing around a patent by making a substantial change, merely makes an insubstantial change, essentially misappropriating or even 'stealing' the patented invention, infringement may lie under the doctrine of equivalents. Application of the doctrine of equivalents is the exception, however, not the rule. . . .

Id. (citations omitted).

"The essence of prosecution history estoppel is that a patentee should not be able to obtain, through the doctrine of equivalents, coverage of subject matter that was relinquished during prosecution *to procure issuance of the patent.*" *Hoganas AB* v. *Dresser Indus., Inc.*, 9 F.3d 948, 951-52 (Fed. Cir. 1993) (*"Hoganas"*) (emphasis added). "The legal standard for determining what subject matter was relinquished is an objective one, measured from the vantage point of what a competitor was reasonably entitled to conclude, from the prosecution history, that the applicant gave up to procure issuance of the patent." *Id.* at 952.

"The doctrine of prosecution history estoppel bars 'a patentee from enforcing its claims against otherwise legally equivalent structures if those structures were

excluded by claim limitations added in order to avoid prior art.' " *Vaupel*, 944 F.2d at 882 (quoting *Mannesmann Demag Corp.* v. *Engineered Metal Prods. Co., Inc.*, 793 F.2d 1279, 1284 (Fed. Cir. 1986)). "In determining whether prosecution history estoppel applies because of a change in claim language during prosecution, the court must consider not only what was changed, but the *reason* for such change." *Id.* (emphasis in original) (because the record does not indicate that the limitation was added to avoid prior art, there was no estoppel to finding infringement by equivalents). "The scope of estoppel must be determined in light of the prior art that occasioned the change, as well as representations made to the patent examiner as to the reason for change." *Sun Studs, Inc,* v. *ATA Equipment Leasing, Inc.*, 872 F.2d 978, 987 (Fed. Cir. 1989) (no estoppel where the discussion of patentability before the PTO was unrelated to the description of the geometric shape) (*"Sun Studs"*), *overruled on other grounds by A.C. Aukerman Co.*, v. *R.L. Chaides Const. Co.*, 960 F.d 1020, 1038-39 (Fed. Cir. 1992). "[E]stoppel is not automatic whenever a claim is cancelled. . . . " *Id.* at 987. Prosecution history estoppel is ordinarily a question of law.

SMC vigorously argues that this Court should adopt the master's recommendation that Festo is estopped under the doctrine of prosecution history estoppel from claiming that it is entitled to a range of equivalents which would erase "meaningful structural and functional limitations of the claim on which the public is entitled to rely in avoiding infringement." *Hoganas*, 9 F.3d at 954 (quoting *Perkin-Elmer Corp.* v. *Westinghouse Elec. Corp.*, 822 F.2d 1528, 1532 (Fed. Cir. 1987)). In *Hoganas*, the court found

prosecution history estoppel where: the examiner during prosecution issued an office action rejecting claims for obviousness; the claimant distinguished the claims so as to relinquish coverage of capillary-sized fibers; and the examiner subsequently allowed the claims based on this distinction. The court held: "In our view, a reasonable competitor could have concluded, from a reading of the prosecution history, that the examiner relied on this distinction in allowing the claims, and thus, that Hoganas had given up coverage of the subject matter incorporating this distinction." *Hoganas*, 9 F.3d at 953.

Here, in the First Office Action, the examiner rejected all claims under 35 U.S.C. § 112 par. 1, which provides: "The specification shall contain a written description of the invention, and of the manner and process of making and using it, in such full, clear, concise, and exact terms as to enable any person skilled in the art to which it pertains, or with which it is most nearly connected, to make and use the same, and shall set forth the best mode contemplated by the inventor of carrying out his invention." The examiner commented: "Exact method of operation unclear. Is device a true motor or magnetic clutch." Claims 4-12 were also rejected under 35 U.S.C. § 112 for being improperly multiply dependent. Report 37. They were not rejected for obviousness under 35 U.S.C. § 103.

The initial claim 1 did not limit the sleeve of the driven member to being constructed of a magnetizable material. Claim 8, which was dependent from claim 1, had such a limitation. Report 31. In response to this first office action, Stoll cancelled independent claim 1 and substituted a completely new independent claim which

required that the driven member have a sleeve constructed of a magnetizable material. Report 32.

Although the examiner identified certain "cited references believed pertinent," there is nothing in the record or the report to indicate that the limitation of a magnetizable sleeve was necessary to distinguish the claimed invention over prior art. Further, there is nothing in the record or the report to suggest that the limitation of a sleeve of magnetizable material was necessary to meet the requirements of specification under 35 U.S.C. § 112. Indeed, the reason why the newly substituted independent claim has the limitation that the sleeve be made of magnetizable material is a mystery, as nothing seems to turn on its composition for purpose of procuring the patent. At the very least, it is a question of fact. *Sun Stud*, 872 F.2d at 987. This Court is not persuaded by SMC's reliance on *Haynes Int'l., Inc.* v. *Jessop Steel Co.*, 8 F.3d 1573, 1577-78 (Fed. Cir. 1993) ("*Haynes*"), in which two claims reciting a relatively broad range of chromium content were abandoned, and a third claim with a more specific chromium content was retained. In *Haynes*, unlike the present case, it was clear from the prosecution history "that chromium content was the principal distinction between the claimed subject matter and the prior art, and this was the primary area of dialogue between Cabot and the PTO." *Id.* at 1578. Accordingly, in light of the circumstances of the present record, this court declines to adopt the recommendation of the master that the claim of infringement by equivalents is barred under the doctrine of prosecution history estoppel.

In addition, with respect to equivalency Festo has presented expert testimony of Dr. Robert Wolf and of Dr.

Stoll to support its claim of infringement under the doctrine of equivalents. Festo argues that SMC has produced no evidence to dispute the testimony of Wolf and Stoll that magnetizable and non-magnetizable sleeves operate in substantially the same manner. However, the patent itself states that if the driven assembly is provided with a sleeve made of magnetizable material, "magnetic leakage fields in the vicinity of the driven assembly can be kept to a minimum. In this way, undesirable breaking forces, which would otherwise be generated while piston and driven assembly pass magnetizable machine components, are avoided." Record 48. This language in the specification, together with the prosecution history, creates a disputed issue of fact appropriate for resolution by a jury under the function/way/result test. *Unique*, 939 F.2d at 1564 (even where there was no literal infringement as a result of plain language of claim and prosecution history, the court examined expert testimony in evaluating whether there was infringement by equivalents). Accordingly, both motions are denied.

3. Literal Infringement of the Reexamined Carroll Patent

Festo has moved for summary judgment on its claim of infringement with respect to the reexamined Carroll patent. Claim 9 of the reexamined Carroll patent requires "a pair of cushion members formed of resilient material, the cushion members being situated near opposite axial ends of the central mounting member to help prevent damage to the piston when the piston contacts an axial end of the cylinder. . . . "

A. 6, 10 and 15 mm slider pistons

The 6, 10 and 15 mm slider pistons have no cushions and Festo does not contend that they infringe.

B. 25, 32, and 40 mm slider pistons

SMC contends that the cushions in the 25, 32 and 40 mm slider pistons cannot prevent damage by making contact with the end plates, because a bolt protrudes from the piston such that the end plate will make contact with the bolt rather than the cushion. (Testimony of Dr. Blanco, Tr. 1800). Festo has neither put forth any evidence to dispute the contention nor pressed the issue in its briefs. Accordingly, SMC's motion for summary judgment on the claim of literal infringement will be granted.

C. Basic Pistons

As to the basic pistons, SMC contends that "in normal operation," their cushions do not "prevent damage to the piston" since they do not make contact with the end plate. This is because "in normal operation" the piston is stopped by stopping the load rather than stopping the load by stopping the piston. The court agrees with the conclusion of the master that the language of claim 9 of the reexamined Carroll patent does not distinguish between situations where contact with the axial end of the cylinder is the normal result of the completion of a stroke and where it comes into play only in the case of decoupling. Report 20-22. Accordingly, the court grant's Festo's motion for summary judgment with respect to the claim of literal infringement of claims 5, 6 and 9 of the

reexamined Carroll patent on the basic pistons with two seals and cushions.

4. Infringement Under Doctrine of Equivalents of Reexamined Carroll and Stoll Patents Concerning Seals

SMC has moved for summary judgment on non-infringement of the reexamined Carroll patent and the Stoll patent based on its use of one seal rodless cylinders, instead of two. Festo has also moved for summary judgment on their claim of infringement by equivalents. Since the Carroll and Stoll patents specifically recite two seals, there is no dispute that the one seal basic and the one seal slider SMC rodless cylinders do not literally infringe.

Claim 1 of the Stoll patent states *inter alia*:

> Said piston further including plural guide ring means encircling said piston body and slidingly engaging said internal wall and first sealing rings located axially outside said guide rings for wiping said internal wall as said piston moves along said tube to thereby cause any impurities that may be present in said tube to be pushed along said tube so that said first annular magnets will be free of interference from said impurities.

Claim 9 of the reexamined Carroll patent specifically defines "a pair of resilient sealing rings situated near opposite axial ends of the central mounting member and engaging the cylinder to provide a fluid-tight seal therewith."

SMC argues that use of only one seal allows dirt to enter the piston chamber and is therefore not equivalent to two seals. The master rejected the argument as "flawed" as applied to the reexamined Carroll patent because it is based on the meaning and function of the sealing rings as described in the Stoll patent, not in the Carroll patent. Report 24. Rather, the Master found that under the Carroll patent: "The seals create a fluid-tight seal against the pressure medium, enabling the piston to be pushed by its force. This results in the piston being efficiently moved by the pressure medium without "blow-by" or air getting around the seals." Report 24. Festo has presented undisputed evidence that for this purpose the one seal in the SMC device performs substantially the same function and achieves substantially the same result in substantially the same way as two seals.

The only evidence which SMC has produced to dispute the claim of infringement by equivalents is statements made by Festo in the context of prosecution of the Stoll patent. In particular, there is a statement by Dr. Stoll that two seals are necessary under the Stoll patent because without two sealing rings, dirt attracted by the magnet will enter the side without the sealing ring. This evidence concerning dirt and impurities does not bear on the meaning and function of the sealing rings as described in the reexamined Carroll patent.

Accordingly, the Court allows Festo's motion for summary judgment that the one seal SMC basic cylinders infringe claims 5, 6 and 9 of the reexamined Carroll patent under the doctrine of equivalents. However, the Court denies Festo's motion for summary judgment that the one seal SMC basic cylinders infringe claim 1 of the

Stoll patent under the doctrine of equivalents. Considering the above statement by Dr. Stoll on the importance of two seals, as well as conflicting testimony from Dr. Wolf and Dr. Stoll at the master's hearing, there is a genuine issue of material fact as to whether one seal functions substantially the same way as two seals in eliminating interference from impurities.

ORDER

For the foregoing reasons, the Court hereby **ORDERS** as follows:

1. SMC is granted summary judgment on non-infringement of the reexamined Carroll patent by the 6, 10 and 15 mm cushionless pistons;

2. SMC is granted summary judgment on non-infringement of the reexamined Carroll patent by the 25, 32, and 40 mm slider pistons with bolts;

3. Festo is granted summary judgment on the infringement of the reexamined Carroll patent by the SMC basic pistons with two seals and two cushions;

4. Festo is granted summary judgment on the infringement of the reexamined Carroll patent by the SMC one-seal rodless cylinders.

5. All other motions for summary judgment are **DENIED** for the reasons stated above and in Court on February 3, 1994.

/s/ Patti B. Saris
PATTI B. SARIS
United States District Judge

TELEFAX

FAX: 6014

TO: Mr. Loesener

FROM: Ralf Haller
FESTO-USA
Fax # 516-435-8026

RE: INK AGITATOR

DATE: Nov. 29, 1988

Achim,

Enclosed is a sketch of ink agitator as discussed on the telephone earlier. As you can see on the agitator sub-assembly, the pinion is prevented from moving off the rack by means of two counter rollers. Also, at the bottom there are 4 rollers going along a guide rail virtually preventing any misalignment of the sub-assembly.

The whole sub-assembly slots into a bracket mounted on the mounting face of the DGO. In addition, the DGO has got play at both ends (about 1 mm) both lateral as well as axial.

The whole agitator is a device sold by our customer to be used to agitate ink, i.e. to prevent skin forming on the surface. Viscosity varies from ink to ink but never approaches that of honey.

As you can see, the possible sideload from the lever on the agitator is taken care of by the 4 rollers at the bottom. Whatever loads are left are minimal and the DGO should be able to cope easily.

The design here incorporates mechanical improvements that I have suggested to the customer during a visit of mine some 2-1/2 months ago. This visit became necessary after a sudden series of breakdowns after very short cycle life to which we could not give satisfactory explanations.

The customer accepted my suggestions and partly redesigned his product. To our embarrassment the situation is still as bad as ever, which we found out just now.

We have had alot of goodwill by the customer but obviously we are now under pressure with very little time and the competition falling over each other on the customer's door step. The account is worth to us $100 K + with a large part being DGO sales.

At this moment in time we have started to conduct tests with the customers device which they have put at our disposal.

Three sample cylinders that have failed in the field after less than 50 miles exhibit very bad seal wear as follows:

1) 2 areas radially opposed by 180 degrees on both seals

2) 3 areas at angles of 120 degrees on the circumference on both seals.

3) 3 areas at angles of 0, 120, and 180 degrees on the circumference.

We also received back an old DGO which we were told by the customer was typical for what they used to have and which at that point in time had done well in excess of 2 million cycles. The seals almost looked new (!!)

I appreciate that at the moment there is not a lot you can do for us but I need to gather as much information as I

can get and there is obviously an outside chance that you may be aware of similar problems elsewhere or you may have access to statistical data that could point in the right direction or that there may be minor but possibly important changes in the specs. or supplier, etc.

I personally still favor looking at an application specific problem, however, in that case we would still need to explain why the cylinders started to fail from a certain point onwards.

At this stage, if there is anything that you feel might be important to help us solve this problem I would appreciate if you let us know.

Best regards,

Ralf Haller

DEFENDANT'S EXHIBIT 66

MINUTES FROM ESS/FESTO MEETING
SEPTEMBER 13, 1988

PRESENT AT MEETING: Tim Epach, Purchasing/ Production Mgr/ESS
William Gardner, Engineering Mgr/ESS
Bernd Ruckdaeschel, Project Engr/ESS
Nancy Novak, Credit Mgr/ESS
Jonn Geragosian, Midwest Regional Mgr/FESTO
Mike Collins, Sales Representative/FESTO

J. Geragosian noted that Ken Hughes has been calling Festo in New York and Charlotte threatening lawsuits and harassing Festo personnel.

T. Epach stated that K. Hughes should not be talking with Festo at this point and should be getting all his information from ESS. He instructed J. Geragosian to have Festo personnel refer Hughes to ESS if he calls again.

J. Geragosian asked about the agitator patent situation; T. Epach explained our license agreement currently in force between ESS and Hughes.

T. Epach gave the background on the test cylinder which failed over the weekend and during last night. He stated that this is a major concern at this point, i.e. whether we can get one to work. He also pointed out that even if we have a cylinder which will work, Festo has so far not provided any data on how long they will last. He stated that we need this information in order to make warranty

decisions, as well as to advise our customers on the life expectancy. He stated that other cylinder manufacturers provide life expectancy ratings for their products.

J. Geragosian stated that Festo's test data shows a cylinder with a 25mm bore will last approximately 1800 miles/stroke, 1200 miles/stroke if run unlubricated. These tests were run with a load of 1/10 linear force at 6 bars pressure. He stated that cycle rates and speeds are major factors in life expectancy, and that you will generally see a 15-25% reduction in life when running on an unlubricated air supply. He noted, however, that no life expectancy claims can be made without extended testing in a particular application.

T. Epach stated that K. Hughes has told ESS that Festo claimed at one point that the cylinders will last indefinitely, but we cannot substantiate who made the statement, if it was made at all.

W. Gardner stated that the original cylinder received from Festo was run 2 million cycles, then cut down to check seals, which were found to be ok. But now, we are having tremendous problems with the cylinders in the field. The original spec called for "clean, dry air." Later, Festo stated that the cylinders were being overlubricated when assembled.

J. Geragosian stated there are two lubrication issues, one concerning the air supply and one concerning the cylinder. On the pre-lube of the piston and the yoke, Festo wants to monitor them carefully to prevent overlubing.

J. Geragosian requested data on which cylinders are failing. It was agreed this would be provided him in approximately two weeks, as it is being gathered now.

W. Gardner stated that one cylinder which was returned was on the press a maximum of 3 months, which calculated out to 115 miles. This cylinder has been sent to the New York office for evaluation.

B. Ruckdaeschel stated that the current speed for the cylinders is approximately 3 inches/second.

W. Gardner stated that the competition is running at approximately 22 seconds for a 40" cylinder, while our speed equates to approximately 15 seconds, which is not totally acceptable to customers.

T. Epach stated that the failures are running approximately 25%. As an example, he stated that a shipment of six cylinders will yield two cylinders which will fail, while the other four operate properly. It appears to be a question of tolerances.

J. Geragosian stated that ESS has an unusual application for the cylinder and that tolerances indeed may be affecting the performance. He also acknowledged that ESS is experiencing seal failures. He said he suspects the problem may be one of not getting a clean valve shift, that air is bleeding out.

T. Epach/B. Ruckdaeschel agreed with this statement.

B. Ruckdaeschel noted problems ESS is having with the flow control valve, and said it appears to be the check valve not operating properly at low cycle speeds; may not be seating correctly.

J. Geragosian will investigate the configuration of the seal.

B. Ruckdaeschel stated Festo may also want to look at a lower micron filter, as dirt appears to be building up around the needle valve, which may be contributing to the switch problems. ESS plans to test a filter of 5 microns.

J. Geragosian suggested taking apart one control valve.

T. Epach noted that we have not had a lot of switch and valve returns of late, and that ESS has worked through the component problems experienced early on. He stated that now we can switch cylinders on a unit and have them work.

J. Geragosian suggested building a fixture with different components to test the new cylinder; suggested using a different limit switch, because he thinks that the current one may not be getting a clear enough signal at the slow speed. He also stated that the new seal is actually a harder seal. This fixture was set up and completed on 9/13.

T. Epach stressed that ESS needs someone of authority from the Engineering Group, and perhaps a Production Manager with Festo to address the problem. He stated that ESS wants to know if we are trying to do the impossible and if we will continue to experience these problems. He noted that we did not see these cylinder problems during the first 6-8 months of production and thus we wonder what has been changed in the cylinder design.

J. Geragosian agreed to investigate the possibility of design changes and will look into getting Engineering personnel in to assist in solving the problem.

A copy of an ESS memo dated September 8, 1988 regarding recent tests run on cylinders was given to J. Geragosian.

J. Geragosian stated that a program could be set up to rebuild cylinders out of warranty, rather than forcing the customer to buy new ones. He also stated that the cylinder is not field repairable.

T. Epach stated that the seals are showing failure problems.

J. Geragosian stated he would like the unit with the new seals to be put on a new fixture with new switches. B. Ruckdaeschel stated that this would be completed today.*

B. Ruckdaeschel stated that he feels that low speeds will create problems, and that he was told by M. Collins that ESS is running at speeds not in the cylinder's design. He asked if this is indeed true. J. Geragosian stated that he could not definitively answer that question. B. Ruckdaeschel stated that he feels we are at the limit of the cylinder's design.

J. Geragosian agreed to become more involved with the situation to ensure it is resolved in a reasonable time. He also stated that he has been talking with Festo/West Germany about whether any changes were made to the cylinder, but not regarding the application. He stated that

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*NOTE: The test fixture was completed on 9/13/88; the unit failed on 9/14/88.

we need to concentrate on the design of the agitator as a whole in order to resolve the problem.

The test cylinder was then reviewed in the Production area.

9/13/88nrn

UNITED STATES DISTRICT COURT
DISTRICT OF MASSACHUSETTS

FESTO CORPORATION,	:	
Plaintiff and Counterclaim Defendant	:	Civil Action No. 88-1814-PBS
V.	:	
SHOKETSU KINZOKU KOGYO KABUSHIKI CO. LTD., a/k/a SMC CORPORATION and SMC PNEUMATICS, INC.,	:	
Defendants and Counterclaim Plaintiffs	:	

Courtroom No. 6
Federal Building
Boston, MA 02109
9:00 a.m., Wednesday
July 6, 1994

Before: THE HONORABLE PATTI B. SARIS, District Judge and a Jury

Second Day of Trial

* * *

[2-19] THE COURT: Right, not for the truth of it. It's just for the marketing techniques that they use.

Go ahead.

(Plaintiff's Exhibit No. 109, Press Release: "New DGO . . . ," received in evidence.)

A. Yes, Exhibition 109 is the typical press release which we send to the magazines and to some of our distributors and employers as introducing the product into the market. Direct mail campaigns, et cetera.

Q. Now, upon introduction of this product, the DGO line, how were your marketing plans directed with regard to this product as opposed to other products of the company?

A. Well, when you come into a market which is completely established as a newcomer, initially we established this Highlights catalog. And it is very difficult to penetrate a new market if you haven't got anything to offer what is not already on the market.

DGO is an innovative product, was a pacesetter for us and was a great door opener when we visited customers. Engineers immediately looked at this. They never saw anything like it, that magnets could be used in a such a configuration, and the features offered great potential and immediately it was a great conversation piece. And we could not only with this product sell this product itself –

[2-20] MR. NEUSTADT: Objection, your Honor. We're again deviating from the question-and-answer format.

THE COURT: Yes. What's the next question?

Q. Mr. Saalbach, has the product sold successfully?

A. The product has been successfully sold until – had continued growth of one hundred to a hundred fifty to 200 percent growth every year until approximately 1988 when it flattened out, and the sales have stagnated equal since then.

Q. And why have they flattened out?

A. Because of the unfair competitiveness in introducing a product by SMC in the market.

Q. Now, is the DGO protected by patents?

A. The DGO has two patents, one patent developed by the principal owner of our company, Dr. Kurt Stoll, and the second patent which was purchased by the parent company from an inventor in order to eliminate any confrontation in the market at that time.

MR. HOFFMANN: Your Honor, I would like to offer into evidence a certified copy of the Stoll Patent, which is U.S. Patent Number 4,354,125. I show it to the –

THE COURT: What number is that?

MR. HOFFMANN: Plaintiff's Exhibit 15.

MR. NEUSTADT: No objection, your Honor.

THE COURT: All right, fine.

* * *

[2-90] THE WITNESS: Okay.

Q. Could you explain the application you're referring to down in Disney World?

A. Yeah. In Disney World we have several applications, but two distinctive ones which I think everyone would identify.

This is Thunder Mountain where this train comes down thrashing this mountain and it's real frightening and you see this waterfall all of a sudden appearing in front of you, and then this big rock falling down. And the rock is actually moving through the water because of the feature of the cylinder. Because it's completely leak proof, it can be used within water. It guides this rock, invisible to the spectator of course, by this device.

The same thing with Twenty Thousand Leagues Under the Sea where you have different exhibits moving as a submarine moves along. Here, again, the same device, the Festo device underneath water, because there is no leakage so you can use it exactly that way.

Q. Are there any other advantages of the magnetically coupled rodless cylinder?

A. Another advantage is because the yoke is only magnetically coupled and not mechanically coupled, it is loose. Consequently, it rotates freely around the stainless steel tube.

* * *

UNITED STATES DISTRICT COURT
DISTRICT OF MASSACHUSETTS

FESTO CORPORATION, Plaintiff and Counterclaim Defendant V. SHOKETSU KINZOKU KOGYO KABUSHIKI CO. LTD., a/k/a SMC CORPORATION and SMC PNEUMATICS, INC., Defendants and Counterclaim Plaintiffs	:	Civil Action No. 88-1814-PBS

Courtroom No. 6
Federal Building
Boston, MA 02109
9:00 a.m., Thursday
July 7, 1994

Before: THE HONORABLE PATTI B. SARIS, District Judge and a Jury

Third Day of Trial

* * *

[3-67] development, was my main activities, production and technical sales, and my brother heads the commercial side.

Q. And have you personally made a number of developments over the years on your own?

A. I made more than a hundred different patents in Germany and worldwide Festo itself has about 600 patents.

Q. And you're talking about patents that cover your inventions?

A. Partly my inventions and partly covering inventions – and partly inventions from other engineers.

Q. Have you been the author of any publications?

A. Yes. The pneumatic field at that time was rather new, so it was very important to give know-how, to give application details to our customers, and that's why we had to be very active in marketing, marketing not only by catalog, price lists, but also by books and by illustrations in trade magazines.

I was the co-author of five different books and about 120 different prints in trade magazines. The books are called in the pneumatic field, first one, Introduction to Pneumatics; second one, Application of Pneumatics; the third one, Pneumatics in the Woodworking Industry; number four, Pneumatics in the Packaging Industry; and the fifth one, Cutting Cost With the Help of Pneumatics.

Besides these books, we have our own trade [3-68] magazine which we send out to 40,000 different customers in 50 countries, and this trade magazine comes

out every four months and I am the publisher of it. It's called Pneumatic Tips.

Q. Now, in regard to the books, what's the scope of publication on the books, the distribution?

A. The first book, the oldest one, called Introduction to Pneumatics, passed the ninth edition just last week, and it's printed in 14 different languages so we can utilize it as a sales help in our 40 different countries, Festo countries.

Q. Have you had anything to do with developing an understanding of terminology in the pneumatics industry?

A. Yes, we found a need on the market that there is no dictionary about special terms in the fluid power industry, and I got a little bit angry about it and started the first 50 pages myself and now today it's on the open – you know, the book shops or whatever, an English/German, German/English dictionary about fluid power, expressions for this industry, and also for schools and universities.

MR. HOFFMANN: Your Honor, I'd like to mark for identification Plaintiff's Exhibit 610.

THE CLERK: Award marked as Plaintiff's Exhibit 610 for identification.

THE COURT: Well, we don't need it for [3-69] identification. Let's not do that. Any objection?

MR. NEUSTADT: Your Honor, I was not provided with the translation of this, so I'm hopeful that someone will tell me what's said here. I have no idea, so

I'd like to wait until after the witness testifies before I state an objection or not.

THE COURT: Well, I don't want – all right, but we're normally not going through the ID route, is that true –

MR. HOFFMANN: Okay.

THE COURT: – because that takes too much court time.

THE CLERK: So I can –

THE COURT: You will, because when it's translated – although that looks like Latin, not German. Is it?

THE WITNESS: That's German. But we can leave it out if you want.

THE CLERK: 610 marked for ID.

(Plaintiff's Exhibit No. 610, Diesel award, marked for identification.)

BY MR. HOFFMANN:

Q. Dr. Stoll, I'm going to show you Exhibit 610 and maybe you can explain what this is and what it says for the Court.

A. I received a gold medal for making inventions. It's [3-70] called the Diesel. Diesel is the diesel engine. And on account of Mr. Diesel a few hundred years ago, the German patent office is handing out three to four medals per year for extraordinary inventions in various fields.

Q. And is this an award that you received?

A. Yes.

Q. And when did you receive it?

A. It was last year.

THE COURT: Wait a minute. So it's not in yet.

MR. HOFFMANN: Okay. I want to offer it in evidence.

MR. NEUSTADT: No objection, your Honor.

THE COURT: Fine.

(Plaintiff's Exhibit No. 610 marked in evidence.)

MR. HOFFMANN: Your Honor, I have two others, award-type documents.

THE COURT: Just show it to him, have him identify it and we'll mark it as an exhibit. Why don't you show it first to Mr. Neustadt.

MR. HOFFMANN: Fine. He's seen them already. Exhibits 592 and 593 shown to the witness.

A. The first one is an honorary doctor's degree from the Technical University in Vienna, Austria, and the second one is a medal received from the German government.

[3-71] Q. And the award from the Technical University in Austria, what is that for?

A. For our – for my achievements in the technical field of fluid power.

Q. And the other award relating to the German government, what is that for?

A. That's also for the same achievements.

MR. HOFFMANN: Your Honor, I'd like to move Mr. Stoll's awards into evidence.

MR. NEUSTADT: No objection.

THE CLERK: University award marked as Plaintiff's Exhibit Number 592, and German Federal Service Cross award marked as Plaintiff's Exhibit Number 593.

(Plaintiff's Exhibit Number 592 marked in evidence.)
(Plaintiff's Exhibit Number 593 marked in evidence.)

BY MR. HOFFMANN:

Q. Dr. Stoll, are you a member of any organizations?

A. Here in the U.S.A. you have a National Fluid Power Association where most of the manufacturers are combined. We have a similar association called VDMA with the same function in Germany, and I was sitting on the board of directors up to last year for the last 15 years.

Q. Are you a member of any other technical organizations?

A. German Engineering Society, VDI, I'm also a member and guest speaker.

[3-72] Q. Now, Dr. Stoll, in the years that you've been with the company since 1957, have your responsibilities grown?

A. At the very beginning where we were a rather small group, I had many more fields of activities as today,

where today I'm concentrating more on technique and organization only.

Q. Is there still a portion of your time involved in the development of new products?

A. I would say at least half of the time is spent for new inventions. Festo is known to bring on to the Hanover exhibition each year at least 30 or 40 inventions.

Q. So since you've been with the company, you've been heavily involved in the development of products, is that correct?

A. Yes.

Q. Now, Dr. Stoll, I'm going to show you a copy of the patent in suit, Plaintiff's Exhibit 15. Do you recognize this patent?

A. Yes.

Q. Is this a patent that covers an invention of yours?

A. Yes.

Q. Could you give us some idea as to how you first became involved with the technology that resulted in this patent invention?

A. Here we are speaking of a linear drive, linear motor [3-73] driven with compressed air.

When we started in the pneumatic field and visiting customers with our first products, we found differently – a need on the market side not only for short-stroke, medium-stroke, but also for long-stroke cylinders, long-stroke in the sense of ten-foot and going even more up to 50-foot. This –

Q. What time period are we talking about?

A. Around in the sixties. There was nothing on the market which could go to such long strokes and I was really tempted to do something in that field. I was looking around what kind of functions could be – could be used for this – for this solution.

And you all know about the mail chute which is also pneumatically driven. You have a tube and inside a capsule which moves back and forward by air pressure, but this kind of application has a lot of limitations, mainly weight and also in volume.

So the basic idea was, why not take a load inside the capsule where you want to transport something, mail or whatever, why not take it outside the tube and take the tube only as a driving means and the load outside of the tube.

So there was your basic idea of the first linear cylinder. And the question came: How can we transform, transmit the interior force, the interior driving energy, to [3-74] the outside, to the load? And from physics we know what we could use a magnetic coupling. If I put a magnet inside and a magnet outside, have a carriage inside, have a carriage outside mounted on rollers, then I could prove that with this prototype that the idea was correct, was working.

Q. Were you aware of anything like that at the time?

A. Pardon?

Q. Were you aware of anything like that at the time?

A. No.

Q. So did you go about trying out this idea?

A. Yes, I built a prototype with a long plastic tubing and used a carriage inside and a carriage outside connected or coupled with a permanent magnet.

Q. And did you actually build this in a room at Festo or how did you go about it?

A. Yes, in our workshop we had made it up and it worked.

Q. Now, are we still talking about 1960's, is that correct?

A. Correct.

Q. Now, when you tried out your idea in this nature, the magnetic coupling in a tube with the carrier on the outside, what did you do when you built it and observed it? Did you take any action?

A. The first action was to apply for a Petty patent in Germany, which we received, to cover the basic idea.

* * *

[3-90] worked, what didn't.

A. We tried on the outside, we tried various material where it's hardness, soft-hard, even up to metal and we found that a semihard wiper action was giving us the best solution for wiping as well as for guiding the outer magnets.

On the inside, we first started to make a soft seal only without the guide. But then the soft seal was bearing down so fast on account of the eccentric action, on account of friction, that I mentioned before, so that the

only solution was to split the two functions, which is leakage and which is a way of – friction, to split the two functions also in two different materials.

MR. HOFFMANN: Maybe it would be helpful if you come over here if it's all right, your Honor, because he's talking about a vacuum as to where he's working in.

THE COURT: Yes.

Q. Now, you were just explaining that you're talking about on the inside. Do you mean the inner piston? Why don't you show us where that is.

A. The inner piston is here.

Q. You said you tried a soft ring first, a soft type of seal ring, is that correct?

A. Right.

Q. What happened when you tried that?

[3-91] A. The soft ring was bearing down relatively fast on account of the eccentric motion.

THE COURT: You know, this is a problem with this huge room. If you'd just standing so that your back isn't totally to me. I just couldn't hear at all what you had to say.

THE WITNESS: Should I stand on this side?

THE COURT: No. That way the jurors don't hear. Just like were you are right now would be fine.

THE WITNESS: Okay. The Piston is here, the soft seal is here, and when we took only a soft seal, then the soft seal would wear down very fast because we have too much friction for the two magnets.

Q. Does that relate to that radial force you were talking about?

A. Off the radial force, correct.

And then we found that we have to split the two functions of sealing against air and of guiding against friction to split these two different functions into two different material –

Q. First before you split it –

A. – a soft material for leakage and a hard material for wear, because both together would be a compromise and would not work.

Q. Okay. Now, before you tried to split them between a [3-92] softer and a harder material, did you try to just make the first ring harder?

A. Yes. We tried to make the first ring harder but then the sealing effect was limited. Because in the tubing we get unevenness, and the unevenness of the tubing of the surface, of the inside surface of the tubing you can seal only if you take a soft enough seal. With a hard seal you won't get a leakage free surface. So that's why we had to go back to a soft seal and the hard seal did not work for these two functions which are necessary to make it movable.

Q. So you said then you tried two different rings.

A. Two different rings and two different materials, a soft material for sealing and a hard material for guiding.

Q. And how did you take care of the dirt problem you were talking about?

A. The dirt problem we have inside and outside. Inside for the dirt which comes with the compressed air and also the dirt which could come from impurities from the steel tubing itself or from the magnet.

On the outside, we have no compressed air, so outside we have to fight only against the dirt from the outside. And that's why we could go here to a harder seal than on the inside because we don't –

THE COURT: Now that you've confused me, are we talking about a harder seal or a harder wiper? See these [3-93] terms are being used –

THE WITNESS: I just wanted to say a harder seal like inside we could not take. We could not take because outside we don't have any compressed air, so it doesn't have to be soft. We went right away to a harder wiper. And the harder wiper has two functions: A wiping function against the dirt and a guiding function against the friction. So by combining these two functions against dirt and friction, we are able to solve the outside portion, because outside we don't have the leakage problem.

Whereas, there are three problems inside. Leakage, friction and dirt we were able to solve with the principle which I mentioned, which is the soft seal and a separate hard seal – guide for guiding. Excuse me.

Did I make it clear?

Q. Right. So I think that's clear enough. And you're able to take care of the three internal problems, sealing, wiping, and guiding –

A. And guiding.

Q. – by the two sets of rings, is that correct?

A. And outside the two solutions.

Q. Okay.

THE COURT: Excuse me. Now, when you say "outside two solutions" –

THE WITNESS: Two functions.

[3-94] THE COURT: Two functions.

THE WITNESS: Two functions, solutions for the two problems; for the problem of wear and the problem of dirt. Outside we don't have the problem of leakage.

THE COURT: That's two wipers?

THE WITNESS: Pardon?

THE COURT: How many wipers?

THE WITNESS: One wiper for each side.

Q. Okay.

Did you eventually reach a satisfactory solution in this matter?

A. Yes.

Q. How long a period of time are we talking about that you tried these different things and materials and rings and solutions to come up with the answer?

A. That's a time consuming process because we wouldn't go to the market with any prototype which is not tested at least for a few million cycles or 10,000 kilometers. That means we go into a torture test so that

we can prove the quality before it goes to the customer. Otherwise, we could not be specified as the first European company by General Motors to become a supplier, so quality is quite a big issue for us.

It's not only a slogan or a nice pamphlet, but for this quality test it took a long time not only for this [3-95] product but for other products to come to the final solution.

Q. Why don't you go back to the stand. I think that will be adequate for that.

A. (Witness complied.)

Q. Did there come a point in time when you applied for a patent when you felt you had a unique design and applied for a patent?

A. After going through all of these details here and getting the final results from the torture test chamber, I found now we had made a big step, a big progress in the right direction and that should be covered by a patent because it's a real step forward for making a lineal motor with a magnetic coupling workable.

Q. And what was the essence of the invention that you felt you had that was in that application?

A. The essence was to solve the three problems, leakage, dirt and friction and the solution of it, the soft seal, the hard guide and the wiping and guiding function.

Q. Okay.

Is that the patent that we're involved with here the Plaintiff's Exhibit?

A. Yes. Claim No. 1 is concentrating on these three functions.

Q. Now, at the time, were you involved directly

* * *

[3-110] A. If I recollect it correctly, it was a little later.

Q. It was a little later?

A. Yes.

Q. Do you know how much later?

A. Not exactly.

Q. Today, it is your belief that radial is better than axial?

A. They both work. Technically they only have the price difference on account of the different production procedure.

Q. We're talking about your patent. Is there any other aspect of your patent that you feel is not necessary in the U. S. Patent?

A. I found a difference in Claim No. 1. I mean, in the main claim, because it reads different in the American Patent than in the seven other countries where we've also got the patent, which does not contain the mentioning of magnetizable sleeve. This wording is not mentioned in claim No. 1 in all other countries that we applied for the patent. Only in the American one this language came in.

Q. And what is your concern about that language?

A. Well, it's not of importance, the magnetizable sleeve, because it doesn't have to do anything with the function, what we mentioned before about friction, about leakage, about dirt. This shielding effect of the magnetizable sleeve is outside and is not even touching all the three [3-111] elements, wiping, guiding and sealing, what we mentioned before.

Q. So what you're saying, in the essence of the invention, you don't believe that the sleeve has any effect?

MR. NEUSTADT: Objection, leading.

THE COURT: Sustained.

A. It has a shielding effect, and we did not specify any material with our claim. We said only if you use.

Q. Okay. Let me go to that.

In the specification you suggest the use of a magnetizable sleeve as a possible portion of the structure, is that correct?

A. Correct.

Q. And why did you put that in?

A. The only function of the magnetizable sleeve is to keep the magnetic flux, the magnetic field inside the lineal motor. That was the only reason for putting it in.

Q. Now, have you found anything out about that feature over the years since you filed the application?

THE COURT: Can I back up for just one second?

THE WITNESS: Yes.

THE COURT: What do you mean by magnetizable sleeve?

THE WITNESS: Magnetizable sleeve, it can be any material which is magnetizable. We did not state any

* * *

[3-115] car. In both functions you need an outer shell.

MR. HOFFMANN: Okay.

THE WITNESS: And it doesn't matter whether we take steel, aluminum or whatsoever.

Q. Now, during times you gave testimony in this case. At one time we were talking about seal rings. How many seal rings do you have on a Festo cylinder?

A. If you take a rodded cylinder, we can work with one seal ring if it acts in both directions.

If you have more space, like here, on the DGO and magnetic tie, we prefer a sealing on each side outside the magnets. A soft seal on this side, a soft seal on this side.

Q. Could you come over again? Maybe it's best to work with your drawing.

A. (Witness complied.)

Q. When you're referring to the piston, you have a seal ring shown. What seal rings are on the piston? I'm talking about the magnetically coupled ones?

A. We have here the seal ring, and the seal rings have to get the air tightness, leakage zero from the compressed air here. So we can take two different kind of seal

rings. A, a seal ring which works from both directions, like an O ring, for instance, or a double seal, or, what we prefer is the single seal, which works only for this direction, or the [3-116] second one, for this direction (Indicating.)

So you can take either two seals or you can take one single seal. They both work, but the double seal has more life expectancy. Because a double seal has a self compensating effect. Even if it wears down, it still is pressed out by air pressure. Whereas, a single seal, either a round or any kind of figure, configuration, if it wears down, it does not have a self-compensate – a way of compensating for that.

So you either can take a single seal for the piston or – which is working in both directions, or you can take two seals which are working in one way and in one way.

THE COURT: And just to be clear, when you are saying two seals, you mean one seal on each end?

THE WITNESS: Right.

THE COURT: Not two seals on each end?

THE WITNESS: No, one seal here and one seal here, on each end.

Q. Now, if you use a two-way seal, which you can use, you said –

A. Right.

Q. – you'd need one on each end?

A. No, only one, only one.

Q. So when you were –

A. A one-way seal here, a one-way seal here, or a double [3-117] seal, two-way seal, whatever you call it, on one side only is sufficient.

Q. Now, you could put two, or four, or six or eight?

A. Any kind, but minimum of one.

Q. Okay.

Now, when you were talking about the Festo structure and using always two, were you talking about the two one-way seals?

A. Two one-way seals.

Q. Why does Festo prefer that in their commercial form?

A. Because you get a higher life expectancy and because you have a redundancy effect. If one breaks down, you still have the other.

Q. Okay.

Now, during that period of questioning, you were asked if there was only one sealing ring, you said dirt could get in. What did you mean by that?

A. The sealing ring acts for leakage, as well as for dirt. So if you would eliminate the sealing ring, you still have the guiding ring which take away the dirt or impurities.

Q. And that would keep it from the magnets?

A. Keep it from the magnets.

Q. Now, if you didn't have a guide ring in there, what would happen?

[3-118] A. Then you would get the dirt to the guide rings, if you don't have a seal ring. So the best mode is from – the minimum mode is two guide rings, one sealing ring and the best mode is two plus two.

Q. Okay.

MR. HOFFMANN: I'm trying to think if there is anything else we haven't covered. I don't think so. I think that's all I have.

THE COURT: Thank you.

Let me just ask you this, Mr. Neustadt. Do you need Dr. Stoll up here or would you rather have him back on the stand?

MR. NEUSTADT: Back on the stand.

THE COURT: Back on the stand. That's great. Why don't you move the chart back.

MR. NEUSTADT: Over there?

THE COURT: Yes.

While we're doing this, why don't you stand and stretch, a final stretch.

CROSS-EXAMINATION BY MR. NEUSTADT

Q. Dr. Stoll, can you see this all right?

A. (Nodes head.)

Q. This is a model that we made. This is the inner member or the piston, and this is the tube and this is what

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UNITED STATES DISTRICT COURT
DISTRICT OF MASSACHUSETTS

FESTO CORPORATION, Plaintiff and Counterclaim Defendant V. SHOKETSU KINZOKU KOGYO KABUSHIKI CO. LTD., a/k/a SCM CORPORATION and SMC PNEUMATICS, INC., Defendants and Counterclaim Plaintiffs	Civil Action No. 88-1814-PBS

Courtroom No. 6
Federal Building
Boston, MA 02109
9:10 a.m., Friday
July 8, 1994

Before: THE HONORABLE PATTI B. SARIS, District Judge and a Jury

Fourth Day of Trial

* * *

[4-20] refers back to that statement there that we read before?

A. Right.

Q. So you do have responsibility for this, do you not?

A. Yes, but I don't have responsibility for claim number 1, which was totally changed. That's what I mentioned before. I don't feel responsible for claim number 1, which is completely different from claim number 1 in German, British/English, seven other countries, and claim number 1 in the American final, converted, improved, changed claim number 1. That was what I was referring to and not to the general description.

Q. Is there any way in which someone reading your patent would know that you do not feel that you're responsible for the language in the patent?

A. I'm responsible for the language in the patent if I know of, but for the changed version, this was not my – this I didn't come across to me. This was signed here. It says here: "Attorney signing the application letter." Here (indicating) are the page numbers, so and so, and what we mentioned just now. On the back page states my signature an don the front page it says: "Attorney signing the application" of the changed version where the number 1 claim, the important claim, was changed without my knowledge, That's what I meant. I cannot be held responsible for a change which I didn't know.

* * *

[4-57] you don't have to differentiate, and at that time we had only radial.

Q. Now, Dr. Stoll, we were discussing yesterday the difference between one seal and two seals?

A. Correct.

Q. Okay. And I think you told me that on all Festo rodless cylinders there are two seals.

A. Preferably.

Q. Pardon me?

Q. Preferably.

Q. Preferably. I'm not asking about preferably. I'm asking about a fact.

A. That's what I mentioned yesterday. You can do it with two or with three.

Q. Yes, but my question is how has Festo done it in the past with its commercial rodless cylinders.

A. We have done it with two single-acting, one-way seals.

Q. Always two, correct?

A. Correct.

Q. Okay. Now, you remember I referred –

A. What I showed you yesterday on my sketch where we were also doing it with one double-acting seal.

Q. Now, you remember the deposition testimony that I referred to previously with Mr. Kelly. This was in Mr. Hoffmann's office?

[4-58] A. I wouldn't recall the details.

Q. And do you remember he too asked you about the use of two seals?

A. Two soft seals.

Q. Well, we'll see. Let me show you what his question and your answer was. This is Mr. Kelly referring to your patent.

"Continuing down column one, line 49, there's reference to a piston which is slidable in the tubular part and then the disclosure goes on to say and which has sealing means at each end for wiping engagement with an internal surface of the tubular part. My question, Mr. Stoll, is, who suggested using two sealing means on the piston?"

And then your answer was: "You always need two when there's a double cylinder, so I proposed it."

And then Mr. Kelly said: "When you say double cylinder, you mean double-acting?"

A. Correct.

Q. And your answer was: "Yes. I suggested it because you always need two when there's double-acting cylinder."

And then Mr. Kelly said: "What will happen if I only use one sealing means?"

And then your response was: "Then on the side where there's no seal, dirt could come in. Attracted by the magnet."

[4-59] And then Mr. Kelly finished up with: "To your knowledge, has Festo ever made a rodless cylinder with only one sealing means?"

And your answer at that time, at least, was: "Always both sides. Always two. One on each side or one on each end."

A. That is correct, and that's what I mentioned yesterday. There are two ways of doing it. There's a double-acting, two-way seal, and there are single-acting seals where you need two, and the life expectancy is much higher in a one-way seal where you need two pieces. I made it quite clear yesterday.

Q. And in this testimony here, didn't you say that you need two seals to keep the dirt out?

A. Yes.

Q. And do you remember in the patent office in the prosecution of your patent telling the patent office the same thing, namely that it was necessary to keep the dirt out?

A. The dirt out? You can do it by the soft seal as well as by the guiding ring.

Q. Well, I'm not asking you how you can do it. I'm asking you if you recall what you told the patent office, and maybe I can help you by showing you exactly what it was that was said in your application.

* * *

[4-61] A. Uh-huh.

Q. And that's what he's talking about when he says: "These references."

He says: "These references," which were the two patents cited by Festo, "are obviously clearly distinguishable over the subject matter of the claims" – those are your claims – "now present in this application."

And then he says: "It is clear that neither of these two references discloses the use of structure preventing the interference by impurities located inside the tube and on the outside of the tube while the arrangement is moved along the tube."

Is your understanding of this statement that the two references did not disclose a mechanism for keeping the dirt out? Is that your understanding of this statement?

A. Impurities? Yes, located inside and outside, correct.

Q. These are the impurities that you said before can be attracted by the magnet.

A. In the sense of dirt. Impurities and dirt is the same, right?

Q. Right.

A. Okay.

Q. So this was the statement, as you understand it, by your attorney to the patent office in your patent application that your invention distinguished over these [4-62] references because you were able to keep the dirt out?

A. Correct.

Q. Okay. And do you recall that you said to Mr. Kelly in that deposition we just read that if you have only one seal, the dirt will come in on the side that doesn't have the seal?

A. Again, the same what I mentioned before, yes and no.

Q. And so isn't it true that if the cylinder that we are examining has only one seal and let's the dirt come in, that it is not part of your invention as recited in this exhibit?

A. Could you repeat that, please?

Q. Let me rephrase it.

In this statement to the patent office, you're saying that your invention is important because it keeps the dirt out.

A. Correct.

Q. And that the other references don't keep the dirt out.

A. Correct.

Q. And in the deposition that we just read with you and Mr. Kelly, you told Mr. Kelly that it you had only one seal, dirt would come in where you did not have a seal, attracted by the magnet?

A. In the one-way seal.

Q. So here you're arguing that your invention distinguishes over the prior art because it keeps the dirt

* * *

[4-100] Incidentally, in this certified copy we have established that it is totally out of order. Papers get mixed up in a patent office and so I'm having difficult.

Q. Well, here is the board.

A. May I go over there?

The examiner issued his office action, which gives the results of his examination. And, essentially, what he said, this is a paper attached to it, it's a summary form. Essentially he allowed the claims, which would have gone to patent, but he had some questions as to formalities, which, again, is a very common thing. His questions as to formalities under certain sections, they set these sections. The sections they're referring to here, 35 U.S. Code 112 say you have to be clear. Remember I told you in the application, you have to present it to the office, you have to present a very clear specification as well as clearly pointing out the claims.

The examiner had a question. "Exact method of operation unclear." And he then specifically simply asked a question. "Is device a true motor or magnetic clutch?" He just said, what is it, one or the other?

Now, when they went in the application, which was translated from the German, it called this whole deal here that we're talking about a linear motor. The examiner had this question on that point. A question simply as to [4-101] semantics.

He also said, "Improperly multiply dependent." He was talking about the claims. You have to specifically number the claims in a certain way. Under a German practice or foreign practice.

THE COURT: Hold on.

(Fire engine going by.)

THE COURT: It's hard enough to hear in this courtroom without outside interference.

(Pause.)

THE COURT: All right.

THE WITNESS: Thank you, your Honor.

A. Under German practice and other countries' practice they can give certain numbers in the dependent claims, referring back to more than one or two. It's easily straightened out – because remember I said it was a translation of the German – by renumbering. So the examiners found no art, no prior art, nothing in his examination that said you can't have a patent because Jones shows this and Smith shows that. He said: I find it allowable. Clear this up, tell me what it is.

Q. Okay.

Now, as a result of that action, was there a response by the applicant's attorney? ?

A. Yes.

[4-102] Incidentally, you have a three-month period in which to respond back. You have the right to respond back to the examiner with respect to this office action. It's called a response. And, again, all of this has to be in writing.

Q. Now, what was contained in the – we have another board here, maybe you can exp'ain what's on this board for the Court and the jury.

A. Should I come there?

Q. Yes. Why don't you move over there.

THE COURT: The one with file history on it is exhibit?

MR. HOFFMANN: The one with the file history the office action on it is P-68, I believe.

A. Correct, Plaintiff's Exhibit P-68.

THE COURT: Do the jurors have that in their little book?

MR. HOFFMANN: Yes.

THE COURT: Okay. Now what are we going to now?

MR. HOFFMANN: 31A which is not in the book but it's the Claim 1 of the issued patent.

THE COURT: Which is Exhibit 14.

MR. HOFFMANN: Right.

THE COURT: Don't they have the issued patent in [4-103] their book?

MR. HOFFMANN: The issued patent's in the book, yes. That's the exhibit right in the beginning, the first exhibit is the issued patent. Claim 1 of the issued patent.

THE COURT: All right, you have all these three documents in your book. You can go back and forth with them when you hit the deliberation room.

Q. Now, why don't you explain what happened or how a response was filed by the applicant's attorney.

A. The exact is in the file history, but this is essentially what happened here. the United States attorney for Dr. Stoll in response to that office action came back and he canceled Claim 1, which is that claim over there,

original Claim 1, he canceled it, which is permissible under the practice; and he canceled some other claims of those original 12, and he inserted a new claim, which was 13.

At the very end of examination as a patent issues they renumber all the claims. It may be confusing when we talk about original Claim 1 and now new Claim 1 which went in as 13. Claim 1 in the patent was this Claim 13 that went in at this point.

When the U. S. attorney for Dr. Stoll put in this Claim 13 that I'm pointing to here, he added additional language to the claim. He also made a very brief statement to the patent office saying: I'm resubmitting this and here [4-104] it is.

He also, at that time canceled or amended the claim – Strike that.

What he did in his claim that he resubmitted, he no longer referred to either a linear motor or a clutch, but he used entirely new words. He used "in an arrangement" and so forth. So in his new claim he did not answer that directly.

Q. In your opinion what did he do? What had he done?

A. Well, he added a lot of unnecessary language.

MR. NEUSTADT: Objection.

THE COURT: Overruled.

Q. Can you point to any particular language that is significant here to what we're talking about?

A. Well, right here the words we've been hearing –

THE COURT: Unless they've got the greatest eyesight in the world . . .

Q. You will have to read it and they have copies of the claim. Count down the lines of the claim perhaps and they can follow –

A. Well, it's not the same as in their book. I'll read it. There is a statement in here. "Wherein said driven member includes a cylindrrical sleeve made of a magnetizable material and encircles said tube," he added this language among a few other things in there which in my opinion –

MR. NEUSTADT: Objection, your Honor, move to [4-105] strike. He's giving testimony now –

THE COURT: Overruled. It's his opinion it's unnecessary language. You can cross on it.

Now, what's the next question?

Q. I would like him to finish the answer.

What is your opinion on adding that language?

A. He added this terminology, he added that term in there which in my opinion is unnecessary. The examiner was willing to allow the claim, as I pointed out over here, he just had a couple of questions here. In my opinion, all that was necessary for the U. S. attorney to say is it's a linear motor, that's what I take it to be, the testimony is it's a linear motor.

As to the second, he simple had to say renumber it. That's why I say this language was unnecessary.

Q. Would your opinion with the claim as originally filed with an explanation as to linear motor would that have been allowed?

A. Yes.

Q. And would that then have appeared in the issued patent as originally filed?

MR. NEUSTADT: Objection, leading.

THE COURT: Sustained.

Q. If that original Claim 1 had be allowed, what would have been the result of that?

[4-106] A. It would have appeared in the issued patent as Claim 1, the independent claim.

Q. Okay.

However, after he submitted the new claims, what was the next step in prosecution?

A. The examiner in effect allowed it at that point with the new claims, but there was in the specification, the written portion that explains – there was still a reference to linear motor, so the U. S. attorney had not changed that or addressed that, so he asked them to clear that up. So the U. S. attorney simply entered what we call amendments and asked the patent office on Line 6 to change "linear motor" to "magnetically coupled arrangement" or whatever term used, and then the patent issue.

Q. As a result of the prosecution history, have you reached any conclusions regarding the scope of the claims?

MR. NEUSTADT: Objection, your Honor.

THE COURT: Sustained.

Q. Are you familiar with the Doctrine of Equivalents?

MR. NEUSTADT: Objection, your Honor.

THE COURT: Let me see you at side bar as to where this is going.

[4-107] (CONFERENCE AT THE BENCH AS FOLLOWS:

THE COURT: What's your question going to be?

MR. HOFFMANN: My question is going to be what is the Doctrine of Equivalents. And follow up: In your opinion, that would prevent the Doctrine of Equivalents from being applied?

THE COURT: I will sustain that objection.

MR. NEUSTADT: A clearly legal conclusion.

THE COURT: What I will allow, however, and we talked about this at length in conference beforehand, because as you know, I should be issuing an opinion soon. This is the only area I really disagreed with the Master on. Mr. Neustadt said that one issue maybe whether or not a reasonable competitor looking at the prosecution history would believe that the inventor was forced to relinquish or abandon a certain limitation in order to procure the patent. I may not have the words exactly right but that's the gist of it from all the cases. I think the Hoganas case is the one I'm thinking of, particularly, the one you cited to me.

MR. NEUSTADT: Yes.

THE COURT: So I would allow him to say that from his experience in the field in doing this would a reasonable person or competitor use this as an abandonment. What I won't allow is questions about: Does this trigger [4-108] prosecution estoppel or Doctrine of Equivalents? That kind of thing.

MR. NEUSTADT: I don't think this witness has an expertise to testify as to his competitor. He is just a patent attorney.

MR. HOFFMANN: I disagree with that. He represents many companies, and that's what a company does, it goes to a patent attorney.

THE COURT: Would somebody familiar with the patent field, I would allow that, would a reasonable person looking at this history with familiarity with the prosecution world conclude that it had to be given up.

MR. NEUSTADT: I'll state an objection for the record.

THE COURT: You can, but I'm assuming that's what you're going to do with your expert.

MR. NEUSTADT: Not with a patent expert.

THE COURT: You're right, but I think the reasonable competitor, he may not know about the reasonable competitor, but he can certainly talk about the reasonable person with patent experience whether looking at this history what they would understand from that. I will allow him to do.

MR. NEUSTADT: Then he's just giving an opinion of law as a patent attorney.

[4-109] THE COURT: No, no. What I won't let him do is give the opinion of law.

MR. HOFFMANN: I have one other question.

THE COURT: I'm not doing all this now.

MR. HOFFMANN: No, just one quick question. You said that he would put part of his case in right away. Part of our case is his cross.

THE COURT: That's right. The only thing that's going to happen in an hour is that I measured it for on best mode. But, otherwise, it's all been cross on infringement. There's about an hour on THYSSEN that I won't debit you for. . . . END OF CONFERENCE AT THE BENCH.)

[4-110] BY MR. HOFFMANN:

Q. In your opinion, would a man skilled –

THE COURT: Person.

Q. – person skilled in –

MR. HOFFMANN: Sorry, your Honor.

Q. – a person skilled in the analysis of the file wrapper history conclude that anything had been given up or dedicated or released from the patented invention from reviewing the file history and comparing what happened, particularly old Claim 1 and issued Claim 1?

MR. NEUSTADT: Let me give my objection, your Honor. I know you've overruled it. That's a legal conclusion.

THE COURT: Please, you need a basis for it. I'm actually going to sustain this on form. I know he's struggling with the form. But, maybe you could just rephrase that about somebody who is experienced in the patent field.

Q. Somebody who is experienced in the patent field, would somebody reviewing the file history, who's experienced in the patent field, is it your opinion that they would conclude that anything had been given up in the difference between original Claim 1 and what was Claim 1 of the finally issued patent?

MR. NEUSTADT: Objection, leading and also the [4-111] other objection, too, would someone conclude that.

THE COURT: It wasn't exactly how we talked about – at the risk of just having to do this 17 times, if I could just see you at side bar.

(CONFERENCE AT THE BENCH AS FOLLOWS:

THE COURT: The problem I'm running into, it's not so much whether something was given up or added. Of course it was. The standard is that they had to relinquish it in order to procure the patent, as I understand the Hoganas statute ruling. But someone who is experienced in the field, looking at this prosecution history, reasonably concluded they were required –

MR. NEUSTADT: Your Honor, that is not the standard.

THE COURT: Well, what do you think the standard is? Because I want to just make sure that this isn't an issue on appeal.

MR. NEUSTADT: It most definitely is. The standard is, it starts with Doctrine of Equivalents is an exception, not the rule.

THE COURT: I know that.

MR. NEUSTADT: You're not limited by the prior art. What he's required to give up –

THE COURT: Please, I don't need the dissertation. [4-112] I've gotten it in the brief. The question is I don't want to go up on this question. As I understand it, it is not just that something was abandoned or changed. Of course it was.

MR. NEUSTADT: The standard is, what does a reasonable competitor conclude from looking at this, not – a reasonable competitor does not say, was he required to give this up. It's just what does a reasonable competitor conclude from reviewing this?

THE COURT: All right. You can phrase it alternatively, if you want. I'm not going to tell you your view of the law is.

MR. HOFFMANN: Would someone experienced in the field be able to determine, would somebody experienced in the field determine whether the applicant was required to relinquish any of the subject matter that differs between issued Claim 1 and originally filed Claim 1?

THE COURT: All right. And any variation that you would like of that. But it can't just be abandoned or changed, because, of course, everyone agrees it was changed.

MR. NEUSTADT: That's right. Required to relinquish.

THE COURT: The reason they did it was in order to procure.

MR. NEUSTADT: Because there is no rule that if you [4-113] only give up what you're required to give up. The rule is –

THE COURT: I tell you what, I know that's your thing, you filed it, and they can phrase it however they want, but it's stupid to just say to give it up and change it. Because, that's –

MR. HOFFMANN: Or were you required to relinquish it?

THE COURT: However you want. I mean, I'm not going to dictate that to you. But the way you said it was –

. . . END OF CONFERENCE AT THE BENCH.)

[4-114] BY MR. HOFFMANN:

Q. Mr. Chovanes, in your opinion, would someone experienced in the field conclude, in reviewing the file history be able to determine whether the applicant here was required to relinquish any of the subject matter that differs from – between issued Claim 1 and originally filed Claim 1?

MR. NEUSTADT: Objection, legal conclusion and objection, leading, and also an objection that this witness doesn't have any particular expertise to rule on this, but I know you are overruling it.

THE COURT: Overruled. And it's obvious you are not somebody in the industry. The question is, would somebody with a background and understanding of patent histories and prosecution histories, would you be able to give an opinion on that one way or another?

THE WITNESS: My answer is yes, that one knowledgeable in this field of patents could render an opinion as to whether or not anything was given up by this –

MR. NEUSTADT: Objection, your Honor. He's now giving the legal conclusion as to what was given up.

THE COURT: Overruled.

I've only gotten half way through the answer, so I'm not sure what he's saying, so I'll listen to it. [4-115] Go ahead. Maybe you can start again.

A. My answer is yes. I believe that one skilled in patents and knowledgeable in patent prosecution, in patent law, could render an opinion in this matter on the question of whether or not anything was given up by the lawyer here when he canceled Claim 1, as originally filed, and inserted Claim 13, which eventually become one in the issued patent.

Q. I believe the question is, could you determine whether there was a requirement that he give it up as well?

A. Additionally, I believe that one could render an opinion as to whether there was a requirement as to whether anything was required to be given up.

Q. And do you have an opinion in this respect?

A. I do.

Q. What is your opinion?

A. My opinion is one, he was required to give nothing up over Claim 1, as originally filed, and, two, in fact, he gave nothing up.

MR. NEUSTADT: Objection, your Honor. Legal conclusion, move to strike.

THE COURT: Yes. I'll strike the part about whether he did give something up or not. That will be one of the key issues for you to decide as you saw on the question form. But I will allow the opinion as to whether somebody looking at this history could reasonably conclude [4-116] that he was required to give it up.

Q. Do you have an opinion as to whether Dr. Stoll was required to include in issued Claim 1, was required by the patent office in the proceeding to conclude – include in issued Claim 1 the requirement of a magnetizable sleeve?

MR. NEUSTADT: Objection.

THE COURT: I'll allow the answer, but only based on your review of the prosecution history and what that tells you, not a legal conclusion.

MR. NEUSTADT: I think he's going to give a legal conclusion.

THE COURT: Well, no, just based on what you've read in the prosecution history, is there anything in that which you see as requiring a limitation of the claim.

Go ahead.

A. My answer to that is that based on what I see from the file history there was no requirement that he add that limitation, he add that language.

Q. Now, was there any requirement – I withdraw that question.

Is it your opinion that there is any requirement in the prosecution history to have the necessity of two sealing rings?

MR. NEUSTADT: Objection. Objection, your Honor. Legal conclusion.

[4-117] THE COURT: I'm not even sure I understand the question. I'm having trouble ruling on it, so I'll sustain the objection as to form.

Q. Okay.

In regard to the number of sealing rings, you're familiar with – you read the patent, you read the specification and you studied the file history.

A. I have.

Q. Did you find any requirement in your review of the file history that the claims must be limited to two sealing rings?

MR. NEUSTADT: Objection, legal conclusion and also as to form.

THE COURT: Let me see counsel at side bar. This is difficult.

[4-118] (CONFERENCE AT THE BENCH AS FOLLOWS:

THE COURT: As I understand it, there is no argument of prosecution history estoppel with respect to the sealing rings. Is that right? It's just a question of whether they meet the graver test. So why are we going into this?

MR. HOFFMANN: Okay. I'll withdraw it.

THE COURT: You've never raised that.

MR. NEUSTADT: I don't think I understand your question, I'm sorry. Maybe I –

THE COURT: The closure prosecution history estoppel goes only to magnetizable sleeve.

MR. HOFFMANN: That's right.

MR. NEUSTADT: I think prosecution history estoppel, as I understand it, also goes to the two seals.

THE COURT: You never raised that, never, ever, before anybody.

MR. NEUSTADT: I think you and I are talking about different things.

THE COURT: You might argue that given the prosecution history, you can help use that in construing claims.

MR. HOFFMANN: He's talking about the equivalency testimony.

THE COURT: You're not using it as an estoppel, [4-119] since you're just using the history for interpretation with respect to the sleeves. Right?

MR. NEUSTADT: What I'm saying –

THE COURT: And I'll let him ask the question. If you're not pressing it, fine.

. . . END OF CONFERENCE AT THE BENCH.)

UNITED STATES DISTRICT COURT
DISTRICT OF MASSACHUSETTS

FESTO CORPORATION, Plaintiff and Counterclaim Defendant V. SHOKETSU KINZOKU KOGYO KABUSHIKI CO. LTD., a/k/a SMC CORPORATION and SMC PNEUMATICS, INC., Defendants and Counterclaim Plaintiffs	Civil Action No. 88-1814-PBS

Courtroom No. 6
Federal Building
Boston, MA 02109
9:15 a.m., Tuesday
July 12, 1994

Before: THE HONORABLE PATTI B. SARIS, District Judge and a Jury

Sixth Day of Trial

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[6-52] (Conference at the bench as follows:

MR. NEUSTADT: I'd like to move for a directed verdict.

THE COURT: Do you have anything in writing?

MR. NEUSTADT: Yes. Well, I have my notes and some cases for you and –

THE COURT: Why don't we reserve. You have timely moved, we will reserve argument for a later point in time, although I've seen so much of this case through the many, many motions for summary judgment, I'm probably not going to grant it, but I want to give you the right to state it on the record and I don't want to take the jury's time now.

MR. NEUSTADT: I can do it in like about – all right.

THE COURT: Want to do it now? Go ahead. But it's got to be a minute because I don't want them sitting there.

MR. NEUSTADT: Just a minute?

THE COURT: Yes, otherwise we will do it later on.

MR. NEUSTADT: I move for directed verdict or judgment as a matter of law, as it's now called, with respect to noninfringement, with respect to the magnetizable sleeve, with respect to the two seals, also with respect as [6-53] to the nonmagnetic tube with respect to their position changes on that.

THE COURT: The nonmagnetic sleeve?

MR. NEUSTADT: There is a magnetizable sleeve and the claim has in it a nonmagnetic tube, and if you adopt one position on one, it has to be consistent. We haven't gotten into it –

THE COURT: You haven't at all. I didn't even get it. I was waiting and I couldn't wait to hear. I didn't hear anything that would create a prosecution estoppel on the magnetic sleeve.

I thought you were getting into that with the prior art, you would show me some reason they had to abandon it. I didn't hear it. I was on the edge of my seat.

MR. NEUSTADT: That's what I wanted to tell you about, your Honor, and that's why I brought these cases up here. This is not really a prosecution history estoppel case.

When you start to analyze whether a patent is infringed, first thing you do is go through every element of the claim, and the federal circuit says every element is material, you cannot ignore an element.

THE COURT: I agree. Can we go by this? I don't want to do this. I've heard it ad nauseam. I've invited you to rebrief it on whether or not attorney error, [6-54] something like that, would create a prosecution history estoppel.

We couldn't find the case. You should be getting it today or tomorrow, my opinion on summary judgment. I thought you would create an – I'm glad you're abandoning the prosecution estoppel.

MR. NEUSTADT: No, your Honor.

THE COURT: We will do it later.

MR. NEUSTADT: I reserve the right to do it later with respect to noninfringement and with respect to best mode.

THE COURT: Okay. But it's been timely filed.

. . . end of conference at the bench.)

* * *

[6-69] supposed improvement.

Can you define that for me?

A One of the improvements of the Stoll claims is magnetizable sleeve surrounding the outside magnets or the follower member.

Q What is the function of that magnetizable sleeve?

A As described in the patent, the function of the magnetizable sleeve is to prevent the influence of external magnetic steels of the function of the system.

Q And you mentioned a second supposed improvement.

What would that be?

A The second supposed improvement consisted of an exclusionary sealing of the piston by using two seals, one at each side of the piston, to prevent dirt from going into the magnets.

Q Do you think your understanding of the Stoll patent would be the same as one of ordinary skill in this art?

A I believe so. I have read the patent and that's what I interpret. That is the only thing a person skilled in the art can go by.

Q The first supposed improvement you mentioned was the shielding function.

Have you had an opportunity to review the SMC cylinders to determine whether or not it has that shielding function?

[6-70] A The SMC cylinder does not have any shielding function in the follower.

Q With respect to the second supposed improvement, the exclusionary function, have you had an opportunity to consider whether or not the SMC rodless cylinders have this exclusionary function?

A No. The SMC rodless cylinder only has one seal so the dirt cannot be excluded from the piston because it enters from the opposite side, so there is no way to protect the cylinder from either dirt or magnetic particles, not by action of any sealing system. It does not have it.

Q With respect to the magnetizable sleeve in the Stoll patent and with respect to the sleeve in the SMC rodless cylinders, is there any difference in structure between them?

A The magnetizable sleeve called for by Stoll in the patent has the function of preventing outside magnetic fields from influencing the action of the cylinder.

Since the SMC does not have that magnetizable sleeve on the outside, it is not protected from outside magnetic fields, therefore, it can be influenced from outside magnetics fields. It doesn't have that structure. The material is different and it does not have the function because it does not influence any effect from magnetic fields from the outside, therefore, it doesn't have the same [6-71] structure, does not have the same function and the results are that the SMC magnetic system is not

protected from outside influences, therefore, it's not material.

Q Now, I'd like to refer to the second supposed improvement, the exclusionary function, and I'd like you to compare the two-seal Stoll patent claim with the one-seal SMC rodless cylinder; and first I wish to ask you if there is any difference in structure between the two?

A There is a difference in structure because one has two seals and the other one has only one. Because of that difference in structure, the SMC that does not have the additional seal; it only has one, it is not protected from the inclusion of particles on the cylinder and the function, therefore, is not the same because the structure is not the same. It does not have that protection against dirt or the seal.

A The result, therefore, is that it is possible that if you use air, which is contaminated anyway, the contamination can reach the seal in the case of the SMC cylinder and not in the case of the Festo cylinder.

Q Now, I would like you to compare the result between the magnetizable sleeve in the Stoll patent and the result obtained by the SMC rodless cylinder.

A Since the SMC follower, the sleeve on the outside, is not protected against magnetic fields, it can experience the

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UNITED STATES DISTRICT COURT
DISTRICT OF MASSACHUSETTS

FESTO CORPORATION, Plaintiff and Counterclaim Defendant, vs: SHOKETSU KINZOKU KOGYO KABUSHIKI CO., LTD., a/k/a SMC CORPORATION and SMC PNEUMATICS, INC., Defendants and Counterclaim Plaintiffs.	Civil Action No. 88-1814-PBS

Courtroom No. 6
Federal Building
Boston, MA 02109
9:20 a.m.,
Wednesday
July 13, 1994

Before: THE HONORABLE PATTI B. SARIS, District Judge and a Jury

Seventh Day of Trial

* * *

[7-76] A We call it a Mardrive transporter.

Q Mardrive transporter. Why do you call it a Mardrive transporter?

A It was made within the Mardrive system so that's where the name came from, and its purpose is to transport articles in the first place, limp fabric, things within a clothing factory, shirts, et cetera.

Q How did the subject matter of your patent come to be?

A Marine Engineering, as I've already said, was very strong in the manufacture of clutch motors for the garment industry.

Q Maybe you can explain what a clutch motor is.

A I think I have already. It's simply a means where the young lady operator can press her feet on a treadle, this causes the sewing machine to run, she removes the pressure from her feet and it stops. It's as simple as that.

We had been making this for quite a long time and we had a fair amount of expertise in it. And a lot of these motors were going into a place called Remploy. I'm not sure that you have anything like it in the United States.

Q Maybe you can explain Remploy?

A It was a government-funded arrangement where people who were in an accident or they are incapacitated in some way are found employment within protected factories.

Q So is it like a handicap facility?

[7-77] A That's right. Well, you know, quite young people, quite old people, but it's intended to get them back into some form of work, where you can get paid.

Q So what were you doing in connection with Remploy?

A We had already supplied them with many thousands of these clutch motors and the Department of Industry had watched the work we were doing to assist these people. A lot of them could not afford to keep pressing the feet on the treadle all day long. They couldn't afford to lift the foot of the sewing machine, couldn't afford to move the hand wheel of the sewing machine into a needle up position.

So we had designed a pneumatic brain which we fitted to the machines which permitted the handicapped people to overcome these disabilities.

Now, the government had seen the work we had been going on for years and they approached Marine Engineering and said we have a specific project. Will you join us in this project on the basis of pound for pound.

Q You mean pound for pound –

A I can't – we get a pound, they get a pound.

Q You mean the currency in England?

A Yes. We were interested in this. It was, I thought at the time, a relatively simple project. We had already put all the automatic functions into the sewing machines. What they wanted to happen, when the lady sewed two pieces of

* * *

[7-87] A Yes. It's part of a catalog that we would send out to prospective purchasers of the Mardrive transporter. It describes generally some of the applications for

what it's used for and it also contains kind of an order description form at the rear.

Q Was the government very happy with this device?

A Yes. They were quite pleased.

Q Did they make a film of the device?

A First of all, they invited the company to apply for what is known as a design award.

In the United Kingdom, annually, all the major companies that have anything new to show enter a competition and it's again organized by a government agency called the Design Council. So they asked us to enter the transporter in this competition, if you like.

And we were extremely lucky because if you got, if a company got a design award, the bait was about 200,000 pounds worth of free publicity. We were quite interested in that. So we applied.

What we didn't know about design is that there was something else, there was a prize, i.e., a surprise, the top prize in the United Kingdom for a designer. This was awarded by His Royal Highness the Duke of Edinburgh. And it was his personal prize, if you won this prize, he offered – I'm sorry – he would permit you to get any British designer [7-88] to make anything you wanted and he personally would pay for it. It was quite unusual. So we were very lucky.

The company won the design award, the company got 200,000 pounds, I got the Duke's personal prize and the government said we should make a film of this and

show it all around the world, made in Briton, some sort of rubbish.

Q What was the purpose of the film?

A It was really to promote English exports. It just filmed in a few minutes in a TV station. Nothing better to do.

MR. BODNER: Your Honor, I have a copy of that film and I'd like to show it now, Plaintiff's Exhibit 32 and I'd like to offer it into evidence.

THE COURT: Have you seen it?

MR. NEUSTADT: Yes. No objection. (Plaintiff's Exhibit No. 32, Videotape, received in evidence.)

THE COURT: Is there talking on this?

MR. BODNER: Yes.

THE COURT: There is no need for the court reporter to take this down, right, because it's an exhibit.

MR. BODNER: No, your Honor.

MR. NEUSTADT: May I have, your Honor, Professor Blanco view this along with us.

THE COURT: Absolutely. Come on up,

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UNITED STATES DISTRICT COURT
DISTRICT OF MASSACHUSETTS

FESTO CORPORATION, Plaintiff and Counterclaim Defendant, vs: SHOKETSU KINZOKU KOGYO KABUSHIKI CO., LTD., a/k/a SMC CORPORATION and SMC PNEUMATICS, INC., Defendants and Counterclaim Plaintiffs.	Civil Action No. 88-1814-PBS

Courtroom No. 6
Federal Building
Boston, MA 02109
9:20 a.m.,
Thursday
July 14, 1994

Before: THE HONORABLE PATTI B. SARIS, District Judge and a Jury

Eighth Day of Trial

* * *

[8-104] that there's no evidence that SMC copied the Festo device.

MR. BODNER: Your Honor, this is not in a wilfulness sense.

MR. HOFFMANN: There was testimony they were all the same products.

MR. BODNER: He admitted that they all look the same.

MR. NEUSTADT: There is no evidence of copying.

THE COURT: The only concerning I have is – you're right, there is no evidence of copying. You're so versed in this area the law, you understand the legal significance of the word copying as opposed to infringing.

I'm not sure whether I'm not going to be flagging the whole thing in a way no one has picked up on. Do you really want a cautionary instruction?

MR. BODNER: We will object to that if that goes in, a cautionary instruction.

MR. HOFFMANN: She is saying –

MR. NEUSTADT: You could put it in the obviousness instruction that you give them, that there has been no copying.

MR. BODNER: Absolutely not. We will object

* * *

[8-119] specification. So I'm going to give you certain factors that you should consider in interpreting the scope of the claim.

The first one is the ordinary meaning of the words used in the claim. Repeating, the ordinary meaning of the words used in the claim. The second one is the specifications and drawings in the patent. The third one is the prosecution history. The prosecution history. The fourth one is the other claims of the patent. And finally, the expert testimony. The expert testimony. Where a disputed term would be understood to have its ordinary meaning by one of skill in the art from the patent and its history, extrinsic evidence that the inventor may have subjectively intended a different meaning should be given little weight.

Now, let me move on to the second step and explain that a little more. The second step is to decide whether each limitation in the properly construed claims is found in the accused device. Liability for infringement requires that an accused device contain every limitation of a claim or its substantial equivalent. Let me repeat that. Liability for infringement requires that an accused device contains every limitation of a claim or its substantial equivalent.

[8-120] Here the claim of infringement is a claim under what you have now heard referred to many times as infringement under the doctrine of equivalence. Infringement of a patent claim can be found by way of the doctrine of equivalence. It is insufficient for proof of infringement under this doctrine of equivalence that the accused devices are equivalent overall to the claimed

invention. Rather, infringement requires that every limitation of a claim be met literally or by substantial equivalent. There can be no infringement if even a single function required by a claim or on an equivalent function is not performed by the SMC rodless cylinders.

So in order to prove infringement by equivalence, Festo must prove by a preponderance of the evidence that the accused SMC device performs substantially the same function as the claimed invention in Stoll, in substantially the same way to achieve substantially the same result. And that is the test that I've set forth for you, so you don't even have to take notes on it, in 1 and 2. The theory behind the doctrine of equivalence is that if two devices do the same work in substantially the same way to accomplish substantially the same result, then they are the same, even though they differ in name, form or shape. Stated

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